The Integrated Resilient Landscape Approach (IRLA)

An operational framework for assessing the resilience of socio-ecological systems, applied to Haiti's Greater South region



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Working paper, version 03.02

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ABSTRACT

The recurrence of climatic, socio-economic, and institutional shocks in Haiti makes resilience assessment a necessary tool for characterizing local capacities to anticipate, absorb, and adapt to disturbances, and for targeting public action. Yet resilience is often invoked in a normatively positive sense, whereas field observations point to configurations of undesirable resilience in which poverty, ecological degradation, institutional weakness, and climatic hazards reinforce one another and stabilize territories in degraded but persistent regimes akin to socio-ecological traps. In such configurations, asset accumulation is necessary but insufficient to trigger desirable transitions; the policy problem is therefore to weaken the resilience of undesirable regimes while strengthening transformative capacities oriented toward ecosystem restoration, economic security, and social cohesion. This article develops the Integrated Resilient Landscape Approach (IRLA), a mixed-method framework for assessing the resilience of socio-ecological systems (SES) at the landscape scale. IRLA combines (i) a multidimensional composite index of ex ante capacities, built from 50 indicators organized into seven dimensions (physical and infrastructural; institutional and governance; environmental and ecological; economic, livelihoods, and food security; social and community; human; cultural, identity-based, and psychological), normalized into eleven ordinal scenarios and weighted through a structured expert consultation, and (ii) a participatory causal analysis based on causal loop diagrams, used to identify lock-in mechanisms, tipping points, and systemic leverage points. The framework was piloted in coastal and mountain landscapes of Haiti's Greater South, using a stratified probabilistic household survey combined with geospatial and ecological field data. IRLA is designed as a strategic diagnostic tool for prioritizing interventions and supporting territorial governance. Its limitations, in particular the absence to date of validation against observed post-shock trajectories and the sensitivity of composite indices to weighting and aggregation choices, are discussed together with avenues for methodological consolidation.

Keywords: resilience; socio-ecological systems; landscape approach; composite index; causal loop diagrams; socio-ecological traps; Haiti

1. Introduction

Interest in resilience has increased following major natural and anthropogenic disasters worldwide, which have highlighted both the vulnerability of human communities (Koliou et al., 2018) and the limitations of existing analytical frameworks in addressing uncertainty and in strengthening the capacity of systems to anticipate, absorb, and respond to disturbances (Folke, 2006). This challenge is particularly acute in Haiti, where multiple hazards interact with watershed degradation, unplanned urbanization, and fragile infrastructure and institutions, resulting in recurrent human and material losses. In this context, resilience is a key determinant of impact mitigation and recovery, in line with the Hyogo and Sendai frameworks (UNISDR, 2005, 2015). Haiti's Greater South, where high hazard exposure coincides with persistent structural vulnerability, is therefore a particularly relevant setting for resilience analysis. Hurricane Melissa (October 2025) provides a salient illustration. Although Melissa made landfall as a Category 5 cyclone in Jamaica before striking Cuba, Haiti remained outside the core of the system and was mainly affected by several days of heavy rainfall. Despite the absence of a direct landfall, the number of fatalities recorded in Haiti (43) was comparable to that recorded in Jamaica (45). In Petit-Goâve, the nocturnal flooding of the La Digue River accounted for most fatalities, reflecting residential settlement on flood-prone riverbanks, riverbed obstruction, inadequate infrastructure, and limited institutional capacity, including delayed warnings and weak evacuation mechanisms. Intense precipitation can in itself cause severe losses through flooding, but this factor alone does not account for the outcomes observed in Haiti: during the same event, the Dominican Republic was exposed to torrential rainfall of comparable magnitude yet recorded a single fatality. This contrast illustrates that disaster-related human losses are shaped less by hazard intensity than by pre-existing structural vulnerability and exposure (Gillespie-Marthaler et al., 2019).

From this perspective, resilience cannot be reduced to a mere capacity for resistance or post-shock adaptation, but must be understood as a capacity for structural risk reduction and for transforming territorial trajectories. This distinction is particularly important in Haiti, where resilience is frequently invoked to glorify the population's ability to endure extreme conditions. As Lyonel Trouillot (2025) emphasizes in his essay *"Blan an di l, mwen di l tou."* ("The foreigner said it, so I said it too"): "Nothing is more painful than hearing a Haitian echo the discourse that praises Haitian 'resilience.' A deadly compliment: the capacity to survive the worst, to adapt to it. One would have preferred a more liberating virtue, and an appreciation that values rupture rather than continuation." In practice, and departing from its common, largely intuitive and weakly formalized usage, the concept of resilience in socio-ecological systems (SES) represents a clear break from traditional engineering approaches focused on robustness and stability in the face of hazards. It is defined as the capacity of interacting ecological and social components, characterized by non-linear dynamics and the existence of thresholds (Allen et al., 2016), to anticipate and absorb disturbances while maintaining their functions, structures, and identity, as well as to self-organize and learn in the face of change and uncertainty (Olsson, 2003), rather than as a return to a single, supposedly stable equilibrium (Folke, 2006; Walker et al., 2004).

Efforts to assess system resilience and to understand the conditions under which systems do or do not shift toward degraded states, using multidisciplinary approaches, have attracted growing interest (Quinlan et al., 2016; Folke et al., 2010; Eason et al., 2016). Two gaps nevertheless persist in the literature reviewed in Section 3. First, most operational frameworks rely on static composite indices that underrepresent feedbacks, thresholds, and transformation processes; second, few frameworks have been designed for data-scarce and institutionally unstable contexts such as Haiti. This study addresses these gaps by proposing the Integrated Resilient Landscape Approach (IRLA), a conceptual and operational framework for analyzing the resilience of Haitian socio-ecological systems at the landscape scale. Structured around seven interdependent dimensions, IRLA enables the identification of adaptive capacities, early warning signals of degradation, and leverage points for intervention. The study develops a composite territorial resilience index to operationalize this framework across landscapes in Haiti's Greater South (Annex

1). By combining quantitative data on the state of structural determinants with participatory analysis of systemic feedback loops, this work provides an analytical foundation to guide public action toward pathways of resilient transformation that are compatible with socially desirable development objectives and ecological sustainability. The remainder of the article is organized as follows. Section 2 clarifies the conceptual foundations of resilience in socio-ecological systems and the notion of undesirable resilience. Section 3 reviews the main families of resilience measurement approaches and their limitations. Section 4 presents the IRLA framework, its sampling design, indicator selection, normalization, weighting, aggregation, and participatory causal analysis. Section 5 discusses the contributions, limitations, and perspectives of the approach.

2. Conceptualizing resilience in socio-ecological systems: analytical framework and theoretical foundations

2.1 Conceptual distinction between vulnerability, robustness, and resilience

Vulnerability refers to the inability to anticipate or respond to disruptive events (Alwang et al., 2001). It is distinct from poverty, which denotes an observable condition without consideration of response capacities (Thomas, 2008; Provitolo, 2009). Vulnerability describes a pre-existing state prior to the hazard (Adger & Kelly, 1999), corresponding to a latent risk linked to a deficit of resources (Cannon et al., 2003). Its analysis relies on the joint assessment of risks and the capabilities of individuals or households. The expression "vulnerability = risk / capabilities" synthesizes this approach (Rousseau, 2007).

Robustness is generally understood as the capacity to withstand a disturbance without change, a notion sometimes referred to as "engineering resilience" (Holling, 1996). This conception differs markedly from resilience understood as the capacity to change and adapt in response to a disturbance (Walker, 2020). A system is considered robust when it is able to absorb a shock without rupture, while maintaining its initial structures, functions, and performance levels (Dragicevic, 2016). Robustness thus aims at the stability of the initial state and is measured by the maximum intensity of disturbance a system can absorb without shifting to a different state (Anderies et al., 2013).

The literature has distinguished several complementary conceptions of resilience, reflecting different analytical focuses, temporalities, and system dynamics. Engineering resilience is defined as the capacity of a system to withstand disturbances and to return rapidly to a previous state while ensuring the continuity of its performance. From this perspective, the literature mobilizes the metaphor of an attraction basin, often represented by a ball at the bottom of a bowl, in which width reflects the system's latitude, depth its strength of attraction, and proximity to the edge its vulnerability (Holling, 1973). As long as disturbances remain limited, the system returns to its initial state without a regime shift, with resilience latitude corresponding to the maximum magnitude of shocks that can be absorbed before a tipping point is reached (Dauphiné & Provitolo, 2007). Ecological resilience refers to the capacity of a system to absorb disturbances and reorganize while maintaining its essential functions, explicitly accounting for internal feedbacks, slow variables, cross-scale interactions, and the risk of regime shifts (Folke, 2006). This conception is rooted in Holling's (1973) foundational definition of resilience, which views it as the ability of a system to absorb disturbances without losing its identity, thereby emphasizing the dynamic and non-stationary nature of ecological systems. Socio-ecological resilience is not limited to a simple logic of resistance to disturbances, but rather aims to reduce the negative impacts of change while supporting pathways of sustainable transformation (Domptail et al., 2013). It relies both on internal system components (Cutter et al., 2008) and on adaptive processes (Valinejad et al., 2022). Among the numerous definitions of socio-ecological systems resilience identified in the literature (seventeen according to Koliou et al., 2018), the most widely accepted refers to the capacity of a system to absorb disturbances, reorganize, and adapt through learning and change, while maintaining its functions, structures, internal feedbacks, and thus its identity (Folke et al., 2010; Walker et al., 2004; Walker & Salt, 2012; Clément et al., 2018). Transformability more specifically refers to the ability to generate a new socio-ecological state when existing conditions become untenable (Westley et al., 2011). Resilience can

therefore be understood as a measure of the magnitude of change that a socio-ecological system can absorb before crossing a tipping point into another type of state in which the system subsequently tends to persist (Walker, 2020), making it a more dynamic concept than robustness or vulnerability (Berkes & Ross, 2013). As defined here, socio-ecological resilience calls for an appropriate scale of observation and measurement, and landscapes, conceived as socio-ecological systems, constitute a particularly relevant unit of analysis.

2.2 The landscape as an analytical unit and a complex adaptive socio-ecological system

Landscape is a polysemous concept, understood both as a material reality and as a social construct (Dérioz, 2008). This dual nature is recognized by the European Landscape Convention, which defines landscape as an area perceived by people, resulting from the interaction between natural and human factors (Council of Europe, 2000; Margarito, 2019). In landscape ecology, this perspective is complemented by a systemic approach that defines landscape as a mosaic of ecosystems structured by a particular configuration of topography, vegetation, land use, and human settlements, ensuring coherence among natural, historical, and cultural processes (Forman & Godron, 1986). Located at intermediate scales, ranging from a few hundred to several thousand square kilometers, landscapes constitute key functional units for the production of ecological, economic, and social services. Their boundaries do not necessarily coincide with administrative or sectoral divisions, which justifies an approach grounded in the natural, productive, and cultural dynamics that structure them (Buck, 2006). This functional dimension has fostered the emergence of landscape approaches as integrative frameworks aimed at reconciling conservation and development, by explicitly articulating the ecological, social, and economic dimensions of territories (Sayer, 2009; Godfray et al., 2010). By nature multidisciplinary, landscape approaches provide an analytical framework for examining both the visible components of landscapes and the systemic relationships and feedbacks that organize their functioning (Cochet, 2011). Landscapes, which thus emerge from co-evolutionary dynamics between human societies and ecosystems, form socio-ecological systems, or bio-cultural systems, resulting from interactions among communities, institutions, economies, biodiversity, and natural cycles (Robert, 2018).

In this continuity, IRLA considers the functional, or cultural-functional, landscape (Wu, 2021), integrating biophysical dynamics and cultural values, as the socio-ecological unit delimiting resilience assessment. This perspective differs from standardized and centralized frameworks based on imported models and scales. For example, certain reforestation projects associated with payments for ecosystem services may destabilize agrarian systems when they ignore existing land uses or informal tenure regimes, leading to income losses, disruptions in fertility transfers, and displacement of pressure toward other sensitive areas. The landscape scale makes it possible to identify diffuse dynamics at the territorial level and to avoid both a static reading of rural societies and assessments limited to performance indicators alone (Bouvard, 2019; Delarue & Cochet, 2011).

Landscape-based resilience analysis thus consists in identifying the components of the landscape, their spatial organization, and the social relations that structure them (Reed et al., 2020). Landscape quality cannot be reduced to its biophysical properties alone, but also incorporates social, perceptual, and identity-related dimensions (Cassatella & Peano, 2011). Research on spatial resilience shows that resilience depends less on the volume of resources than on their spatial organization, with connectivity and modularity shaping the propagation of disturbances and transformation trajectories (Cumming, 2011). Landscape thus appears as a dynamic ensemble in which interactions among components generate effects that cannot be reduced to the sum of their parts. Its analysis therefore requires a theoretical framework capable of accounting for these dynamics, their non-linearities, and the feedback mechanisms that condition system stability or transformation trajectories.

Building on the framework proposed by Ostrom, landscapes can be understood as complex adaptive systems that bring together resources, actors, governance arrangements, interactions, and outcomes, within which human and natural subsystems interact (Ostrom, 2009). These co-evolving systems ensure the production of essential goods and services, yet their equilibrium

remains fragile and may be disrupted beyond certain thresholds under the effect of crises (Décamps, 2007), leading to transitions toward ecologically and socially degraded regimes (Ursino, 2019). This susceptibility stems from the fact that systems have a limited capacity to absorb disturbances, self-organize, learn, and reorganize without crossing a threshold that leads to an alternative state with different structural and functional properties (Resilience Alliance, 2010). These capacities, which constitute resilience, are intrinsic properties of complex systems, and their assessment relies on the analysis of system dynamics and the systemic interactions that structure landscape functioning (Lagadeuc & Chenorkian, 2009; May et al., 2008).

These dynamics can take contrasting forms. They manifest as systemic trade-offs when improvement in one dimension is accompanied by deterioration in another, for example when increased income from charcoal production intensifies deforestation. Conversely, synergies may emerge when strengthening one component simultaneously enhances other dimensions, such as when biodiversity restoration contributes to food security. Compensation mechanisms may also arise when temporary improvement in one dimension masks the degradation of another, for instance when higher agricultural production temporarily offsets declining fisheries income. Finally, reinforcing feedback loops develop when initial disturbances fuel their own amplification, as in the case of mangrove overexploitation leading to declining fisheries resources and increased pressure on mangrove forests.

These mechanisms may unfold in causal chains producing domino effects, for example when prolonged water stress reduces agricultural yields, exacerbates food insecurity, stimulates internal migration, and weakens social cohesion. In such configurations, systems may reach tipping points: as long as a critical threshold is not crossed, some absorptive capacity remains, but an additional marginal disturbance can trigger abrupt and difficult-to-reverse transformations, such as the critical loss of forest cover (Lenton et al., 2019). These phenomena illustrate the non-linear nature of landscape systems, in which small disturbances can generate major systemic effects. The adaptive cycle and the concept of panarchy, developed by Holling and further elaborated by Folke, therefore invite analysis of landscapes as nested sets of cycles of growth, conservation, release, and reorganization operating at multiple scales, linked by processes of memory and revolt that condition renewal capacities following disturbance (Folke, 2006).

Finally, disturbances differ in their origin, whether internal or external, and in their dynamics, whether sudden or gradual (Aschan-Leygonie, 2000). Atypical exogenous shocks abruptly reveal latent vulnerabilities, while slow transformations progressively alter system operating conditions, as illustrated by rice trade reforms in Haiti in the 1990s, which weakened the national value chain and increased dependence on imports. From this perspective, poverty appears less as a simple deficit of resources than as the product of resilient configurations of social, economic, and environmental interactions that sustain vicious circles and vulnerability traps in which the system has stabilized.

2.3 Undesirable resilience, socio-ecological traps, and asset accumulation

Resilience is frequently mobilized as a normatively positive objective, which constitutes an oversimplification. "Cruel dictatorships, salinized landscapes, or even psychotic states in individuals can be highly resilient. The challenge in such cases is how to reduce their resilience" (Walker, 2020). Within socio-ecological systems (SES), there therefore exist more or less desirable resilient regimes, separated by critical thresholds beyond which feedback loops reorganize and stabilize alternative states. Some of these states may be durably resilient while remaining undesirable, such as poverty traps or self-reinforcing regimes of environmental degradation. A system may retain a high capacity for absorption while remaining locked into an unsustainable regime. In this context, strengthening resilience without questioning the nature of the state being sustained risks consolidating structural lock-ins and slowing transitions toward more sustainable trajectories (Bitterman et al., 2016). The challenge is thus not to increase resilience per se, but to strengthen the resilience of desirable states, as well as the capacities for absorption, mitigation, and transformation that make it possible to avoid or move beyond degraded regimes (Kerner & Thomas, 2014). While poverty traps do not manifest uniformly across low-income countries

and many experience positive growth trajectories (Barrett & Carter, 2013; Kronenberg et al., 2016), when they do take hold they are characterized by critical thresholds below which dynamics become self-reinforcing. Monetary poverty, public underinvestment, low productivity and overexploitation of natural resources mutually reinforce one another, while cash transfers, although providing immediate relief, remain insufficient in the absence of asset accumulation that would allow these thresholds to be crossed.

This analytical framework applies directly to the Haitian case, where poverty is both widespread and structural. Nearly 60 percent of the population lives below the poverty line and 24 percent in extreme poverty, with a GDP per capita of USD 1,272, a Human Development Index of 0.535, a life expectancy of 64 years, and a Gini coefficient of 0.61. Poverty is also understood through deprivation lines related to access to food, water, education, health, and security (Symonds, 2011). While these indicators provide an initial description, they remain insufficient to capture the structural drivers of poverty, notably unequal access to land, colonial legacies, center–periphery relations, and historical forms of discrimination. When the effects of poverty become the causes of its own reproduction, poverty acquires a systemic character, persisting as an emergent state produced by complex dynamics and negative feedbacks (Gore, 2003).

In Haiti, the articulation of these feedbacks corresponds to a socio-ecological trap rooted in long-term dynamics. Decades of massive deforestation, land fragmentation, and resource exploitation without regeneration have progressively eroded the ecological base of the agropastoral system. Crossing critical thresholds, particularly in terms of forest cover and hydrological regulation, has led to generalized erosion, disruption of biogeochemical cycles, and a loss of rebound capacity in the face of climatic shocks. From the 1980s and 1990s onward, trade liberalization and the disintegration of rural value chains further amplified these vulnerabilities, with the collapse of cash crops, the withdrawal of productive investment, and the externalization of economic flows. The reduction of tariffs on rice strengthened dependence on imports and undermined food sovereignty. Declining incomes and the absence of alternatives fostered extractive survival strategies, intensifying ecological degradation. At the same time, state weakness, the erosion of peasant organizations, and the fragmentation of international aid created an institutional vacuum conducive to short-termism, reinforced by uncertainty and aid dependence. These dynamics have stabilized into a degraded socio-ecological regime, characteristic of undesirable resilience. Exiting this state requires activating positive feedback loops, reducing structural dependencies, and promoting asset accumulation beyond regime-shift thresholds, a prerequisite for genuinely transformative resilience.

Breaking out of such an undesirable resilient regime therefore hinges on the ability to cross the thresholds that sustain it. Asset accumulation constitutes a central lever for initiating sustainable pathways out of poverty, beyond mere increases in income or consumption (Ikegami et al., 2017). This logic is formalized by the concept of Micawber thresholds, which refer to critical levels of capital below which households follow trajectories of decumulation, whereas above these thresholds they may enter cumulative processes of accumulation that can lift them beyond the monetary poverty line (Lipton, 1993; Carter & Barrett, 2006). The challenge is therefore not to increase income flows per se, but to enable households to cross these thresholds through multidimensional capitalization strategies, as well as through transformations of the structures that condition access to opportunities (Santos, 2009). This implies coordinated interventions combining capitalization, market access (Ghatak, 2015), safety nets, and institutional and power reforms that structure resource allocation (Barrett et al., 2016). This framework differs from so-called "big push" approaches, whose effectiveness remains limited in the absence of profound structural transformations (Kraay & Raddatz, 2007).

However, asset accumulation remains fragile as long as the systemic thresholds structuring socio-ecological dynamics are not surpassed, due to the persistence of negative feedback loops (Sindzingre, 2007; Pugliese et al., 2017). Indeed, the one-off crossing of a critical threshold does not guarantee the stability of achieved gains. In the Haitian case, unfavorable integration into the global economy reinforces this vulnerability, as the country is exposed to price volatility, competition

from subsidized systems, and trade constraints without fully benefiting from globalization (Gore, 2003). Understanding resilience therefore requires identifying accumulated assets and their trajectories in the face of disturbances, as shocks affect different forms of capital in distinct ways depending on their nature and temporal dynamics (Li et al., 2020). Assessment must thus integrate critical thresholds, feedbacks, and non-linear dynamics at the landscape scale. It is precisely this complexity, stemming from the plurality of shocks, the uncertainty of trajectories, and the systemic nature of the processes involved, that raises the question of how landscape resilience can be meaningfully measured.

3. Measuring landscape resilience: from theoretical concept to operational tool

3.1 Analytical frameworks, scales, and measurement approaches

The transition from the concept of resilience to its operationalization first requires clarifying what is to be measured, at what scale, and under which analytical assumptions. This clarification notably leads to distinguishing the classical opposition between specified resilience and general resilience. The former addresses the question "resilience of what to what" (Carpenter et al., 2001) by focusing on a given system component exposed to a defined disturbance, thereby allowing the identification of control variables, their critical thresholds, and tipping-point risks. By contrast, general resilience refers to the capacity of a complex system to cope with multiple or emerging disturbances, based on structural attributes whose progressive erosion brings the system closer to critical thresholds; analysis then aims to track the evolution of these attributes and the margins of adaptation. Approaches to resilience assessment may also be organized according to both how resilience is defined, objective when it relies on an external theoretical framework, or subjective when it is co-constructed with local actors, and the mode of evaluation, whether based on observable indicators or on subjective and perceptual assessments (Jones, 2019). The literature highlights the absence of a consensual operational definition of general resilience and the strong heterogeneity of disciplinary approaches, linked to the polysemous and hypernymic nature of the concept (Rufat, 2018; Longépée, 2016). While its quantification appears useful for natural resource management, measuring resilience remains challenging due to its abstract, evolving, and composite character, as well as to mechanisms that are still only partially understood. Landscape resilience assessment therefore remains methodologically weakly harmonized and marked by the lack of an integrated model capable of jointly representing physical and socio-economic dynamics across the full disturbance–recovery cycle (Clément et al., 2018).

As an emergent and not directly observable property of socio-ecological systems, resilience is commonly approached through the identification of structural and functional attributes that reflect its underlying mechanisms and allow it to be operationalized. The literature offers a wide range of attribute-based frameworks, each emphasizing particular dimensions of resilience according to specific theoretical and disciplinary perspectives, while collectively contributing to a plural and complementary understanding of the concept. Some approaches adopt a complex systems lens, focusing on latitude, resistance, precariousness, and panarchy in order to capture critical thresholds, shock absorption capacity, and cross-scale interactions (Woloszyn et al., 2013). Others adopt a more functional and operational perspective, emphasizing stability, redundancy, diversity, modularity, preparedness, learning capacity, adaptability, and transformation, in line with planning and management objectives (Kerner & Thomas, 2014). A further body of work foregrounds the social and community foundations of resilience, highlighting the role of social networks, economic diversity, governance, leadership, place attachment, values, and learning capacities in shaping adaptive potential (Berkes & Ross, 2013). Integrative socio-ecological frameworks seek to bridge these perspectives by combining social, institutional, and ecological dynamics, including governance, beliefs, infrastructure, skills, learning processes, and the capacity to accept change (Walker et al., 2004; Walker & Salt, 2006). More recent process-oriented approaches conceptualize resilience as a dynamic continuum of persistence, recovery, adaptation, and transformation, while explicitly accounting for trade-offs among subsystems, system openness, social and institutional

connectivity, participation, and collective learning (Ungar, 2018). In parallel, more synthetic formulations developed in the field of disaster risk studies distill resilience into a limited set of core principles, notably robustness, rapidity, redundancy, and resourcefulness, commonly referred to as the "4 Rs" (Bruneau et al., 2003). Taken together, these perspectives allow the identification of key levers for structural transformation (Koffi, 2014). Despite this diversity, empirical studies consistently identify a relatively stable core of recurring attributes, including functional diversity, redundancy, managed connectivity, small-scale social or productive units, local participation, self-organization, economic autonomy, mutual aid networks, intergenerational transmission, equitable access to information, and capacities for learning and innovation (Dauphiné & Provitolo, 2007; Ungar, 2018; Li et al., 2020; Bernard de Raymond, 2023; Walker, 2020). This convergence supports the view of resilience not as a fixed state, but as an evolving configuration of interacting capacities that changes over time and context (Dauphiné & Provitolo, 2007). To facilitate analysis within the IRLA framework, these attributes are grouped into three complementary capacities referred to as the AAA framework: anticipation, corresponding to the ability to detect disturbances and prepare responses; attenuation (or absorption), referring to the capacity to limit the impacts of shocks; and adaptation, capturing the ability to adjust and, where necessary, durably transform the system. Each indicator in Annex 2 is mapped to one or more of these capacities.

The key challenge then becomes translating these attributes into observable and comparable variables capable of capturing general resilience. It is within this perspective that approaches based on structural proxies and composite indices have developed, constituting the first major family of tools used for the empirical assessment of resilience.

3.2 Approaching general resilience through structural proxies and composite indices

Resilience can be approached through two complementary categories of indicators. Structural indicators describe the ex ante capacities of a system's attributes. They correspond to a formative assessment of adaptive potential, independent of the specific nature of future disturbances, and are therefore understood as proxies for potential resilience. By contrast, dynamic indicators aim to measure realized resilience through the speed and quality of recovery after a shock, for example the return to employment, the reopening of schools, or the restoration of essential services. Their interpretation remains delicate, however, insofar as these trajectories reflect both the internal capacities of the system and the intensity of the disturbances experienced, particularly when rapid shocks combine with slow-onset dynamics (Sharifi, 2016).

For this reason, the majority of methodological frameworks addressing general resilience prioritize the analysis of ex ante capacities, mobilizing structural proxies that capture capital endowments and potentially mobilizable adaptive capacities (Bennett et al., 2005; Dauphiné & Provitolo, 2007). Analysis thus focuses on the structural attributes underpinning system robustness, adaptability, and transformability, which are used as proxies for general resilience (Rufat, 2018). From this perspective, community resilience is defined as the capacity of a community to develop and mobilize its resources to cope with uncertainty and disturbances (Magis, 2010), leading to a focus on resources that can be activated prior to the occurrence of a shock.

These capacities include, in particular, access to land tenure, education level, cash transfers, social capital, infrastructure robustness, institutional preparedness, savings, livelihood diversification, and the adoption of more robust production systems (Rousseau, 2007; Sherrieb et al., 2010). The corresponding indicators are constructed from primary or secondary data and then normalized using common statistical methods (Tan, 2021). Taken together, these indicators provide a composite reading of landscapes, revealing their capacity to absorb disturbances, maintain functional structures, and regenerate after crises (Walker et al., 2002; Décamps, 2007). The identification of resilience indicators therefore relies on the search for observable proxies of robustness, adaptability, and transformability, translated into quantifiable metrics to enable rigorous empirical operationalization (Koffi, 2014; Dauphiné & Provitolo, 2007). This approach is inductive in nature, based on the identification and measurement of a set of characteristics deemed

relevant for resilience, rather than on direct observation of post-shock performance (Winderl, 2014).

This logic has led to the development of a broad family of methodological frameworks based on multidimensional indicators and composite indices, relying on normalization procedures (z-scores or min–max), weighting schemes based on expert judgment or statistical methods, and aggregation into scores enabling inter-territorial comparison and vulnerability mapping (Tan, 2021). These approaches are widely mobilized by international organizations in the fields of territorial planning and climate finance (Rufat, 2018).

Among archetypal frameworks are the Social Vulnerability Index (SoVI; Cutter et al., 2003), the Baseline Resilience Indicators for Communities (BRIC; Cutter et al., 2010, 2014), the Resilience Inference Measurement (RIM; Lam et al., 2016), the Agroecosystem Resilience Index (ARI; Suárez-Pardo et al., 2022), and the UNDRR Disaster Resilience Scorecard for Cities (UNDRR, 2017), as well as several tools developed by development agencies, such as UNDP's Community-Based Resilience Assessment, GOAL's ARC-D Toolkit (McPaul & Mitisidou, 2016), and FAO's Resilience Index Measurement and Analysis (RIMA-II; FAO, 2016). Other studies have also proposed weighted combinations of categories and variables applicable to socio-ecological systems (Kerner & Thomas, 2014; Cushman & McGarigal, 2019). Oxfam's tool measures resilience using binary indicators that verify, for each household, the presence or absence of key capacities enabling shock absorption and adaptation (Oxfam GB, 2013). The CDR model proposed by Deng et al. (2022) conceptualizes resilience as a weighted sum of variables aggregated using a multicriteria method of the TOPSIS type. The ARC-D Toolkit represents a participatory and mixed approach, based on a qualitative scorecard assessing resilience through thirty components linked to the Sendai Framework, collectively scored on a scale from 1 to 5 (McPaul & Mitisidou, 2016). Other frameworks developed by development agencies include the BRACED programmes, combining objective indicators and qualitative analysis; Oxfam's MEAL systems; CARE's Resilience Markers; and the REAL approach, which combines ex ante and ex post measurements (Mellak et al., 2022). In the agricultural domain, FAO's TAPE tool illustrates this logic of ex ante composite indices applied to production systems, through thirty-seven indicators scored on descriptive scales and then aggregated to produce a synthetic profile of agroecological performance (FAO, 2019). A complementary approach is proposed by the Multidimensional Vulnerability Index of the United Nations High-Level Panel, which mobilizes indicators to characterize structural vulnerabilities in a manner comparable at the international scale (UNDESA, 2024).

Overall, these methods facilitate inter-territorial comparison, vulnerability mapping, and the analysis of structural capacities, but they remain largely top-down and tend to underrepresent feedbacks, temporal dynamics, and transformation processes, which are difficult to capture through static indicators (Reghezza-Zitt et al., 2012; Mayunga & Peacock, 2010). This limitation justifies their articulation with methodological frameworks more strongly oriented toward processes, interactions, and temporality, which constitute the families of approaches examined in the following sections.

3.3 Causal, statistical, and macroeconomic approaches to post-shock trajectories

When longitudinal data are available, resilience analysis can rely on statistical approaches aimed at identifying the determinants of post-shock trajectories (Li et al., 2020). These methods primarily mobilize regression analyses, panel models, time-series approaches, or vector autoregressive models to estimate the effects of shocks on economic and social variables. Some frameworks use composite indices as an entry point for more in-depth causal analyses, as illustrated by FAO's RIMA II methodology, which combines objective data with structural models to estimate households' latent capacity to cope with shocks and to identify structural determinants of resilience, such as access to land or education (FAO, 2016). RIMA has also evolved to incorporate a subjective resilience module, illustrating a progressive hybridization of approaches (FAO, 2016). Despite their contributions, these approaches remain constrained by predominantly linear and static representations, which imperfectly capture the feedbacks, non-linearities, and threshold dynamics characteristic of socio-ecological systems. It is

precisely learning, reorganization, and adaptive capacities that enable the opening of new trajectories when existing regimes become untenable (Folke, 2006), yet these dimensions remain difficult to capture through conventional statistical frameworks. To overcome these limitations, probabilistic and reliability-based approaches originating in engineering explicitly integrate a temporal dimension by modeling failure and recovery trajectories following shocks.

3.4 Probabilistic approaches to resilience: reliability, performance, and Bayesian learning

Probabilistic methods originating in engineering conceptualize resilience as a function of performance over time, measured through the cumulative loss of functionality between the occurrence of a shock and the return to an acceptable level of service, formalized by the "resilience triangle" (Bruneau et al., 2003). This framework has been extended to the assessment of critical infrastructure through the generalization of performance curves to different technical systems (Cimellaro et al., 2010) and through the development of specific indices integrating fragility, reliability, and repair times for water, electricity, and transport networks (Todini, 2000; Ouyang & Dueñas-Osorio, 2012; Chang & Shinozuka, 2004). These approaches have also been applied to residential buildings, notably through probabilistic models combining fragility curves, cost functions, and repair times to estimate the contribution of housing to community resilience and to prioritize strengthening strategies (Vona et al., 2018). More recent developments introduce Monte Carlo–based simulation approaches, defining resilience as the probability that a system reaches target performance thresholds following a shock (Chang & Shinozuka, 2004).

In addition to these classical probabilistic approaches, Bayesian models extend reliability frameworks by incorporating progressive learning and the updating of probabilities based on new or partial data. They adopt a probabilistic treatment of uncertainty by estimating the likelihood of future states from incomplete or noisy information, without directly simulating individual behaviors. Their temporal extension into Bayesian networks makes it possible to quantify resilience as the probability of maintaining functionality over time despite disturbances, to identify key variables through sensitivity analyses, and to integrate adaptive learning processes (Franco-Gaviria et al., 2022).

The main strength of these probabilistic methods lies in their explicit and reproducible mathematical framework, which enables scenario comparison and the evaluation of strengthening measures. However, they remain largely focused on infrastructure systems, conceptualize resilience primarily as a return to a reference state, and give limited consideration to social and institutional dimensions, as well as to the adaptive and transformative processes characteristic of socio-ecological systems. While these methods are dynamic with respect to temporal changes in performance, they remain largely static in terms of structural change and learning and transformation mechanisms.

3.5 Systemic modeling approaches: representing feedbacks, thresholds, and adaptive behaviors

Modeling approaches rely on systemic representations that simulate system evolution under different scenarios using mathematical and physical formalisms, integrating interactions, feedbacks, thresholds, and non-linear dynamics (Linkov & Kott, 2019). They require substantial data availability and a detailed understanding of social, ecological, and institutional dimensions. Different schools of thought can be distinguished according to how behaviors and interactions are represented.

- Dynamic models. System dynamics, developed by Forrester (1961), is widely used in ecology and sustainability science. It relies on the representation of aggregated variables, flows, feedback loops, and non-linear relationships. Dynamic models simulate alternative scenarios to analyze system trajectories in response to a policy, a shock, or a given pressure (Filatova et al., 2013). However, these models require long time series, and poor parameterization may lead to misleading results.
- Agent-based models. Agent-based models represent households, firms, or institutions as heterogeneous agents endowed with bounded rationality, learning capacities, local interactions, and spatial embeddedness, allowing complex dynamics to emerge (Filatova

et al., 2013; Valinejad et al., 2022; Donges & Barfuss, 2017). In these models, interactions among agents are formalized through equations or decision rules linking internal states, perceptions, and actions, and updated iteratively. Particularly well suited to socio-ecological systems, they simultaneously integrate diversity, connectivity, and social learning into system resilience. Their potential, however, remains underexploited. Applications are often limited to single indicators, a single type of disturbance, or local case studies that are difficult to generalize, and they rarely include explicit analyses of resilience mechanisms or multi-level governance structures (Egli et al., 2019; Lippe et al., 2019). Their development faces strong constraints related to data availability, computational requirements, and parameterization, as well as still simplified representations of decision-making and cognitive processes, with behaviors frequently modeled through fixed rules (Filatova et al., 2013). Applications such as ResilUS or the Centerville Testbed nevertheless illustrate their ability to simulate post-shock recovery scenarios, while also highlighting their predictive limitations due to behavioral uncertainty and data gaps (Miles & Chang, 2011; Ellingwood et al., 2016; James & Bradshaw, 2023). Agent-based models should therefore still be considered exploratory rather than predictive tools, requiring substantial calibration and validation efforts before operational use, and their policy impact remains limited due to insufficient integration and co-construction (Bitterman et al., 2016; Schlüter et al., 2012; Nugroho & Uehara, 2023).

- Hybrid models. Some models combine system dynamics and agent-based approaches, making it possible to reveal time lags between social decisions and biophysical responses, as well as the decisive role of intangible factors such as fear, cooperation, or past experience in post-shock trajectories (Martin & Schlüter, 2015; Valinejad et al., 2022).

- Input–output and CGE models. In parallel, input–output models propagate shocks through sectoral matrices to estimate direct and indirect economic losses, while computable general equilibrium (CGE) models simulate adjustments of agents constrained by scarcity and their equilibrium transitions after a shock, with some variants integrating temporal and spatial dimensions (Okuyama, 2007; Rose & Liao, 2005).

- Spatial foresight models. Exploratory land-use and landscape-change models based on scenarios combine multivariate analyses and spatial simulations to assess the effects of contrasting policies on landscape configurations. They enable anticipation of territorial transformation trajectories and support planning through prospective mapping. However, these models generally do not represent socio-ecological feedbacks (De Aranzabal et al., 2009).

- Early warning signal models. A final category of approaches analyzes time series to detect a progressive loss of resilience as a tipping point approaches, without explicitly simulating the system. Based on the principle of critical slowing down, these methods identify—sometimes even from short samples (Delecroix et al., 2024)—lengthening recovery times following disturbances through increases in variance and autocorrelation (Scheffer et al., 2009).

Despite their sophistication, modeling approaches remain partially blind to the subjective, social, and institutional dimensions that condition resilience trajectories. To capture these difficult-to-quantify dimensions, a third family of approaches has emerged, based on the analysis of perceptions, lived experiences, and co-construction processes. While modeling progressively integrates social dimensions, it still struggles to fully represent feedbacks and adaptive behaviors (Schlüter et al., 2012). Agent-based models offer a promising avenue for analyzing the emergence of resilience from interactions between structures and behaviors, but remain constrained by strong data requirements and modeling assumptions. Qualitative approaches help to complement these limitations and improve the appropriation of results.

3.6 Qualitative, participatory, and socio-psychological approaches

Qualitative and participatory approaches conceptualize resilience through perceptions, freeing themselves from constraints related to the

availability of quantitative data. Based on surveys, interviews, focus groups, and participatory diagnostics, they draw on tools from the social sciences to produce a contextualized understanding of communities' capacities to cope with shocks, inequalities in access to resources, and possibilities for collective action (Sheller & León, 2016). They thus make it possible to capture risk perceptions, trust, collective norms, and local institutional tensions (UNU-IAS et al., 2024). However, their analytical scope is limited by challenges of generalizability (Reghezza-Zitt et al., 2012).

Several methodological frameworks illustrate this family of approaches, including SHARP+, the COMDEKS program of the Satoyama Initiative (UNDP, 2018), and tools developed by the Resilience Alliance (Martínez-Fernández et al., 2021). The Rural Resilience Index relies on self-assessment by local actors, who evaluate resilience using descriptive indicators scored on Likert-type scales, without mechanical weighting or aggregation; the objective is less comparability than dialogue and the co-construction of action plans (Cox & Hamlen, 2015). In the same vein, the *Indicators of Resilience in Socio-Ecological Production Landscapes and Seascapes*, developed by UNU-IAS and Bioversity International, propose a qualitative approach based on participatory workshops and collective discussion exercises, aimed at social learning and self-diagnosis rather than inter-territorial comparison (UNU-IAS et al., 2024). The Community-Based Resilience Analysis also follows this perspective, documenting, through interviews and focus groups, situated resilience trajectories and the factors perceived by communities as structurally significant (Macopiyo, 2018).

Taken together, these approaches emphasize that resilience cannot be reduced to an aggregate of material endowments. Research in cultural psychology highlights the role of narratives, rituals, kinship networks, and endogenous institutions in rebuilding social ties and supporting psychological recovery after major crises (Ionescu et al., 2010). These approaches help identify the potential of interaction networks and locally perceived thresholds, and should be viewed as complementary to indicator-based quantitative approaches. In contexts such as Haiti's Greater South, this implies considering family solidarities, associative networks, and religious practices not as a mere background, but as integral components of territorial adaptive capacities. This need to articulate structural capacities, social dynamics, and local perceptions calls for integrated analytical frameworks capable of linking measurable indicators with socio-ecological processes.

4. The Integrated Resilient Landscape Approach (IRLA): an operational framework

4.1 A multidimensional analytical framework based on seven key dimensions

The structure is directly inspired by the main families of components identified in international work on disaster resilience measurement, notably the UNDP stocktaking report *Disaster resilience measurements: Stocktaking of ongoing efforts in developing systems for measuring resilience* (Winderl, 2014). In line with this reference framework, IRLA operationalizes the following seven dimensions:

1. The **physical and infrastructural** dimension assesses the robustness, redundancy, and accessibility of essential infrastructures, such as transport, water, energy, telecommunications networks, and collective services.
2. The **institutional, technological, and governance** dimension goes beyond the formal existence of bodies, plans, or regulatory frameworks to capture the quality of relationships among public authorities, local governments, community organizations, NGOs, and the private sector. Institutional adaptive capacity is strongly shaped by levels of trust or mistrust structuring these interactions. Excessive mistrust fosters blockages, conflicts, rule non-compliance, and risk-averse organizational cultures (Stern & Baird, 2015).
3. The **environmental and ecological** dimension distinguishes three complementary families of indicators. Spatial structure metrics describe patch shape, size, and dispersion, informing exposure to edge effects and isolation. Numerical indices characterize the richness, diversity, and internal balance of the landscape mosaic, as well as the dominance of specific elements. Finally, functional indicators assess ecological connectivity, the presence of

redundant pathways for biological flows, and the capacity of the landscape to maintain its essential functions (Finotto, 2011).

4. The **economic, livelihoods, and food security** dimension captures the diversification of productive activities, access to economic resources, and income stability.
5. The **social and community** dimension refers to the core components of social capital, including social relations, reciprocity networks, shared norms and values, trust, collective participation, and access to resources. Three forms of social relations are distinguished: bonding (intra-community ties), bridging (inter-community ties), and linking (interactions between communities and public institutions) (Kirmayer et al., 2009).
6. The **human** dimension reflects available human resources and their adaptive capacity, notably through education, health, and access to essential services.
7. The **cultural, identity-based, and psychological** dimension captures intangible levers of collective mobilization, such as place attachment, trust, beliefs, self-efficacy, and local knowledge. Culture, understood as a set of shared values, practices, meaning systems, and knowledge, constitutes a central resource for resilience by enabling communities to interpret adversity, maintain identity continuity, and sustain collective well-being (Kirmayer et al., 2009).

From a formal standpoint, the unit of analysis $u \in U$ is defined, depending on the context, as a household or a landscape grid cell. The seven resilience dimensions are indexed by $d \in \{1, \ldots, 7\}$. Each dimension d comprises a set I_d of n_d elementary indicators, and the observed (raw) value of indicator $i \in I_d$ for unit u is denoted $x_i(u)$. This notation is used consistently in the normalization, weighting, and aggregation steps below.

4.2 Units of analysis, data sources, and sampling design

The construction of indicators relies on the combination of satellite data, field-based ecological data, and socio-economic data collected from households. Biophysical and landscape dimensions are documented using satellite imagery and geospatial databases, while ecological characteristics, such as relative richness, habitat composition, or the status of specific biological groups, are documented through field observations and measurements. Socio-economic and human dimensions are based on the collection of primary data with statistical representativeness. To this end, IRLA employs a stratified probabilistic sampling design applied to households within the pilot landscapes. The population is divided into strata defined by landscape type (coastal or mountainous), age of the respondent (15–30 years or over 30 years), sex (male or female), and economic category (extreme poverty, poverty, non-poverty). The latter criterion reflects the fact that the majority of households remain in a situation of monetary poverty: in 2012, 37 percent of the population lived in extreme poverty, below 41.6 HTG per person per day, and 45 percent in moderate poverty, below 81.7 HTG, with only 18 percent considered non-poor (World Bank & ONPES, 2014). The combination of these criteria ensures a sample representative of socio-demographic structures. The minimum sample size per communal section is derived from the standard formula for estimating a proportion in a large population, $n = z^2 p(1 - p)/e^2$, under the conservative assumption $p = 0.5$ in the absence of prior distributional information; with a 90 percent confidence level ($z = 1.645$) and a margin of error $e = 7.5$ percent, this yields approximately 120 households per communal section. Proportional allocation across strata ensures that each subgroup contributes to the results according to its demographic weight and enables intergroup comparisons while limiting sampling bias. To reduce selection bias and ensure spatial coverage, households are drawn by stratified random sampling from a georeferenced building database derived from OpenStreetMap. [Insert here: number of communal sections and landscapes covered, total number of households surveyed, survey period, response rate, and the ethics and informed-consent procedure.]

4.3 Selection of ex ante capacity indicators

The IRLA methodology assesses general resilience according to an objective definition and a mixed mode of evaluation (Jones, 2019). It considers resilience as both a structural property, observable in the resources that can be mobilized prior to a shock, and a dynamic property of socio-ecological

systems. IRLA belongs to the family of indicator-based approaches aimed at operationalizing resilience (Suárez-Pardo et al., 2022), while distinguishing itself through the combination of an ex ante assessment of structural capacities and a participatory analysis of feedbacks. Designed as a strategic diagnostic tool, IRLA seeks to measure a state of resilience in order to identify structural fragilities and prioritize transformation levers, without resorting to trajectory simulation. IRLA translates a theoretical framework into measurable dimensions by mobilizing Ostrom's (2009) principles (Mestre, 2014) through heterogeneous indicators. Each indicator is interpreted as a form of capital contributing to the functional mechanisms of resilience and is weighted according to a contextualized judgment before being aggregated into a composite index that enables inter-landscape comparison and action prioritization. The methodology has been applied in several pilot landscapes of Haiti's Greater South, notably in nine coastal municipalities of the Sud department, characterized by high exposure to climatic hazards and pronounced structural vulnerabilities (Annex 1).

As discussed in Section 2.1, resilience is understood as the capacity of individuals and systems to mobilize their capabilities to anticipate, absorb, and adapt to shocks (Walker et al., 2004; Folke et al., 2010). This implies that resilience analysis is analytically grounded in the study of structural social vulnerabilities, which shape patterns of risk, exposure, and differentiated disaster impacts (Cannon et al., 2003), but also condition the extent to which available capacities can be effectively mobilized. In this perspective, resilience cannot be inferred from capacities alone, as their activation depends on both the presence of enabling resources and the constraints imposed by structural fragilities. To address this, IRLA mobilizes three complementary types of indicators. Structural capacity and resource indicators capture the ex ante potential for response, reflecting the assets and enabling conditions that can be mobilized in the face of disturbances. System failure outcomes are used as proxies of structural fragility, providing contextual information on the pressures that limit or distort the effective deployment of these capacities. Finally, indicators of adaptive social dynamics capture the mechanisms through which capacities are translated into action, including collective organization, mutual aid, learning processes, and forward-looking behavior. Together, these three categories allow resilience to be assessed as a relational process linking potential, constraint, and activation, rather than as a static stock of resources. This architecture ensures that the composite index reflects not only the availability of capacities but also the socio-ecological conditions that shape their mobilization and transformation into effective adaptive responses.

Building on this conceptual framework, landscape resilience cannot be measured through a single indicator. Given that landscapes are complex, multidimensional, and highly context-specific systems, they must be decomposed into analytical dimensions while preserving an overall systemic perspective (Cassatella & Peano, 2011). Indicators are therefore necessarily multiple, interdisciplinary, and dependent on the biophysical, social, and institutional context under consideration (Peano et al., 2011). Their selection must meet several criteria, including representativeness with respect to the observed phenomena, data accessibility and reliability, and operational usefulness for guiding public action, while remaining intelligible and usable by administrations and local partners (Bottero, 2011).

The literature highlights the absence of a common list of resilience indicators, due to the limited number of studies empirically testing the links between indicators and actual recovery capacities of socio-ecological systems. Each discipline and agency mobilizes its own metrics, making cross-case and cross-country comparisons difficult (Jordan & Javernick-Will, 2012). While certain variables exert a decisive influence on system resilience, their identification relies on a reasoned selection based on a cross analysis of the scientific literature (Voghera, 2011), expert consultations, and field surveys. This process leads to the combination of biophysical indicators, land use and habitat fragmentation metrics, indicators of agrarian heritage and historical land occupation, anthropogenic pressure, use value, and territorial attractiveness, in order to capture the plurality of functions and values associated with rural and coastal landscapes (Cassatella & Voghera, 2011). An analysis conducted by Argonne National Laboratory for FEMA shows the existence of a core set of approximately twenty sociodemographic indicators recurrently used in

community resilience approaches (Edgemon et al., 2018). Another review covering 73 methodologies identified 1,089 community resilience indicators, which were subsequently organized by capital systems and reduced to a corpus of 98 non-redundant indicators, primarily related to education, employment, income, mobility, household structure, access to health services, housing type, and social capital (Gillespie-Marthaler et al., 2019). This proliferation of indicators makes selection particularly complex for both researchers and practitioners (Koliou et al., 2018; Sharifi, 2016).

IRLA adopts an ad hoc strategy for selecting a core set of indicators based on values expressed by local populations and experts, while retaining common categories that enable international comparison (Annex 2). Within this framework, structural capacity indicators form the backbone of the ex ante assessment, while system failure outcomes provide contextual information on structural fragilities, and adaptive social dynamics indicators capture the effective activation of capacities in practice. This approach aims to produce metrics that are comprehensive in their coverage of system dimensions, directly usable by local actors to inform decision-making, and scientifically robust, grounded in comparable empirical data (Johansen et al., 2016). It also responds to the thematic fragmentation of existing metrics (Li et al., 2024; Johansen et al., 2016). Similar to the Multidimensional Resilience Index developed by CERED in Madagascar, which integrates seven dimensions and combines participatory data with statistical weighting to produce a graduated score (Salava et al., 2021), IRLA proposes a synthetic multidimensional resilience index based on quantitative indicators aligned with the Sustainable Development Goals and other international reference frameworks, complemented, where necessary, by objective proxies. Sensitive dimensions or those prone to declarative bias are inferred from behavioral items developed through community consultations and aggregated into composite scores, allowing the integration of less tangible dimensions such as institutional trust, place attachment, or self-efficacy, while limiting social desirability bias. Adopting a logic of controlled modularity, whereby the enrichment or simplification of metrics does not undermine the conceptual coherence of the index but rather adjusts its analytical resolution, IRLA is based on a set of 50 indicators organized into seven complementary dimensions and derived from approximately 200 underlying metrics.

The number of metrics may evolve depending on analytical objectives and data constraints. The collected observations are then integrated into a normalization and aggregation process, described in the following section, leading to the construction of a composite index.

4.4 Normalization and discretization into eleven reference scenarios: empirical-comparative or locally normative benchmarks

The scores assigned to each indicator are discretized into 11 ordinal classes ranging from 0 to 10. This scale expresses the relative position of the unit of analysis with respect to conditions that are more or less favorable to resilience, rather than a direct measurement of resilience itself (Annex 3). As resilience is a systemic and emergent property, no single indicator taken in isolation can meaningfully characterize it (Cutter et al., 2014; Quinlan et al., 2016). For example, a high level of access to sanitation services is not sufficient to infer health resilience, which also depends on factors such as water quality, urban density, or service governance. Resilience can therefore only be approached through multidimensional aggregation (Quinlan et al., 2016). In the literature, indicator normalization generally relies on a min–max transformation that converts a raw value into a continuous score between 0 and 1, based on the minimum and maximum values observed or defined for each indicator. This standardization makes it possible to compare indicators expressed in heterogeneous units and to integrate them into a composite index. Within the IRLA framework, two complementary logics of normalization and discretization are applied, depending on the availability and robustness of reference data. When reliable empirical distributions exist at the international or regional scale, observed values are positioned relative to the global distribution, in line with benchmarking approaches commonly used in vulnerability and resilience analysis (Lam et al., 2016; UNDRR, 2017). Quantile-based discretization is then applied, with the ten class thresholds defined from the empirical cumulative distribution function of the indicator in the reference sample. The observed value is converted

into an ordinal score reflecting its relative position within this distribution, independently of the original units of measurement, and providing a coherent basis for subsequent aggregation within a composite index.

Formally, let $x_i(u)$ denote the raw value of indicator i for unit $u \in U$, F_i the empirical cumulative distribution function of indicator i in the reference (international or regional) sample, and $K = 11$ the number of ordinal classes, ranging from 0 to 10. The ten interior class thresholds $q_{i,k}$ are defined from the quantiles of F_i, with open outer bounds $q_{i,0} = -\infty$ and $q_{i,11} = +\infty$:

$$q_{i,k} = F_i^{-1}\left(\frac{k}{11}\right), k = 1, \ldots, 10 \quad (1)$$

Each unit is then assigned to the ordinal class $c_i(u)$ corresponding to its position within the reference distribution:

$$c_i(u) = k \; if \; q_{i,k} \le x_i(u) < q_{i,k+1}, k \in \{0, \ldots, 10\} \quad (2)$$

The indicator score used in the aggregation is $S_i(u) = c_i(u)$, expressed on the 0–10 scale; a continuous normalized score ranging from 0 to 1 can be obtained as:

$$s_i(u) = \frac{c_i(u)}{10} \in [0,1] \quad (3)$$

When the assessed attribute is highly context-specific and lacks a robust comparative reference framework, discretization is based on a local normative benchmark defining scenarios considered optimal and critical for the socio-ecological system under study (UNU-IAS et al., 2024). These two scenarios constitute the bounds of the scale, denoted x_i^{min} (critical) and x_i^{max} (optimal); the interval between them is subdivided into ten equal-width intervals, yielding eleven ordinal scenarios from 0 to 10:

$$c_i(u) = round\left(10 \cdot \frac{x_i(u) - x_i^{min}}{x_i^{max} - x_i^{min}}\right) \quad (4)$$

where round(·) denotes rounding to the nearest integer and values outside the $[x_i^{min}, x_i^{max}]$ range are truncated to the bounds. A high score indicates a condition considered relatively favorable or optimal within the local context, whereas a low score signals a condition likely to increase vulnerability.

This discretization makes it possible to compare the relative performance of territories independently of the measurement units specific to each indicator. By assigning an ordinal value, assimilable to a scenario, to each metric across all dimensions, aggregation enables the construction and monitoring of a composite index that constitutes a landscape resilience score, following a logic comparable to that of the Landscape Measures Framework (Buck, 2006; Sayer et al., 2013). Other approaches also rely on idealized scenarios aimed at quantifying how close a territory is to a state considered optimal, whether contextualized or theoretical. The TOPSIS method, for example, measures the distance of a unit from an "ideal scenario" defined by the best performance observed in the sample (Zhao et al., 2025). IRLA follows this logic of relative resilience by constructing a discrete scale of scenarios. Unfavorable indicators, such as unemployment, are normalized through inversion, $c_i(u) \leftarrow 10 - c_i(u)$, so that the highest score always represents the most desirable configuration. It should be recalled that the resulting score reflects a contributory capacity to resilience rather than resilience itself. IRLA scores should therefore be interpreted as an assessment of the quality of the structural foundations from which resilience may emerge, rather than as a direct qualification of a state of resilience.

4.5 Weighting: definition and validation of weights through structured expert consultation

In order to reflect the relative importance of indicators in the Haitian context, weighting coefficients are defined through a structured consultation of a panel of experts, following a co-construction approach that ensures the territorial anchoring of the measurement framework. The panel comprised [n = __] experts from [public administration, academia, civil society, and international organizations], consulted in [__ round(s)] between [dates]. Experts were asked to rate the importance of each indicator for landscape resilience on a five-point Likert scale ranging from 1 ("not important at all") to 5 ("essential"). For each indicator i, the mean expert score, denoted $\bar{x}_i$, was calculated; the dispersion of judgments was assessed through [the standard deviation of scores and Kendall's coefficient of concordance W = __], and indicators with strongly divergent ratings were discussed and re-rated during validation workshops.

Because a score of 1 represents the absence of importance rather than the lowest positive level of importance, mean scores are first transformed using the following quadratic function:

$$w_i = \left(\bar{x}_i - 1\right)^2 \tag{5}$$

where w_i is the preliminary importance weight assigned to indicator i and $\bar{x}_i$ the mean Likert score assigned by the expert panel.

This transformation assigns a theoretical weight of zero to an indicator unanimously considered not important, while preserving the ranking established by the experts. The quadratic function was selected under the assumption that the perceived importance of indicators does not increase in a strictly linear manner. Rather, differences in expert judgments become progressively more meaningful at the upper end of the scale. In particular, the distinction between indicators considered *very important* and those regarded as *essential* was assumed to represent a stronger prioritization than the distinction between lower response categories. Consequently, the quadratic transformation increases the discriminative power of the weighting system by assigning proportionally greater importance to the highest-rated indicators while maintaining the original ordering of expert preferences. This choice is a modeling assumption; its influence on the results is examined in the sensitivity analysis discussed in Section 5.

To facilitate interpretation, the transformed weights are subsequently expressed relative to the smallest strictly positive transformed weight observed across all indicators:

$$W_i = \frac{w_i}{\min\limits_j w_j}, w_j > 0 \tag{6}$$

where W_i is the relative importance weight of indicator i and $\min_j w_j$ the smallest strictly positive transformed weight; an indicator with $w_i = 0$ would be excluded from the index.

Under this approach, the indicator receiving the lowest transformed weight is assigned a relative weight of 1, while all remaining indicators are expressed as multiples of this reference value. For example, a relative weight of 3 indicates that an indicator contributes three times more to the composite index than the indicator receiving the lowest transformed importance weight.

The proposed transformation is not intended to estimate an absolute measure of importance. Rather, it provides a transparent, reproducible and operational weighting scheme that preserves expert rankings while strengthening the differentiation among highly prioritized indicators. This approach therefore reflects both the ranking and the relative intensity of expert judgments regarding the contribution of each indicator to landscape resilience.

The resilience score of each dimension d for unit u is then calculated as the weighted arithmetic mean of the normalized indicator scores:

$$D_d(u) = \frac{\sum\limits_{i \in I_d} W_i \cdot S_i(u)}{\sum\limits_{i \in I_d} W_i} \tag{7}$$

where $D_d(u)$ is the resilience score of dimension d for unit u, $S_i(u)$ the normalized score of indicator i for unit u on the 0–10 scale, W_i the relative importance weight of indicator i, and I_d the set of the n_d indicators belonging to dimension d.

This formulation ensures that each dimension remains expressed on a standardized 0–10 scale while allowing indicators judged to be more critical by the expert panel to contribute proportionally more to the dimension score. The overall landscape resilience index for unit u is then obtained as the unweighted mean of the seven dimension scores, each dimension being considered equally important a priori:

$$IRLA(u) = \frac{1}{7}\sum_{d=1}^{7} D_d(u) \tag{8}$$

The weighted arithmetic mean is fully compensatory: a low score on one indicator can be offset by high scores on others within the same dimension, and a weak dimension by stronger ones at the index level. This property is acceptable for a diagnostic tool designed to compare relative structural foundations, but it should be kept in mind when interpreting aggregated scores; partially compensatory alternatives such as the geometric mean are discussed in Section 5.

The analysis can be conducted at different levels of aggregation, either at the level of households, or through groupings defined by age, sex, municipality, or landscape. This analytical flexibility constitutes a central lever for identifying pockets of fragility, contrasted resilience profiles,

and priority targets for intervention. More specific groupings can also be constructed depending on the objectives of the analysis, for example to isolate women living in extreme poverty in mountainous areas. In this case, profile P is defined as a subset of units sharing the same socio-spatial characteristics, and the associated level of resilience is estimated as the mean of the IRLA(*u*) values calculated over all units belonging to this profile. Denoting U_P the set of units belonging to profile *P* and N_P their number, the average score of profile *P* and its dimension scores are defined as: This approach enables the production of differentiated diagnostics that are directly operational.

$$IRLA_P = \frac{1}{N_P} \sum_{u \in U_P} IRLA(u) \quad (9)$$

$$D_d(P) = \frac{1}{N_P} \sum_{u \in U_P} D_d(u) \quad (10)$$

Because the sampling design is probabilistic and stratified with proportional allocation, the population structure is preserved in the sample: each mutually exclusive stratum, defined by the combination of landscape type, sex, age, and economic category, is represented according to its weight in the population. Under complete response and correctly estimated stratum sizes, simple averages of the observed IRLA(*u*) values within each stratum are therefore approximately unbiased estimators of the corresponding population values, and no additional weighting is required; where non-response or deviations from proportional allocation occur, post-stratification weights should be applied, and standard errors should be reported with the estimates. It would be possible to define a threshold on the aggregated index value, for example one third of the scale as in the Baseline Resilience Indicators for Communities (BRIC; Cutter et al., 2010), in order to identify households or landscapes classified as "resilient." However, such an approach is risky, given the absence of a universal threshold that would allow one to conclude that a territory is resilient based on a single index value (Béné et al., 2012). Interpretation and performance monitoring therefore prioritize visual representation tools. Radar charts, or spider diagrams (Annex 4), make it possible to compare resilience profiles by theme, metric, dimension, or stratification variable, as illustrated, among others, by Sayer et al. (2017) or FAO through TAPE (2019). These analyses can be complemented by choropleth maps using ordered color scales to identify spatial gradients and "hotspots" of vulnerability and resilience.

4.6 Systemic dynamics, participatory causal diagnosis, and co-construction of leverage points

In response to the critique of the "black box" of governance formulated by Biesbroek et al. (2017), and in a context where methods integrating interdependencies and community-scale resilience remain insufficiently developed (Koliou et al., 2018), IRLA mobilizes causal loop diagrams (CLDs; Sterman, 2001) to make explicit the causal chains and feedbacks that structure territorial trajectories. This approach makes it possible to identify genuine governance levers beyond descriptive correlations and symptom-focused responses (Biesbroek et al., 2017). The literature shows that vulnerabilities observed in socio-ecological systems (SES) rarely stem from isolated deficits, but rather from circular dynamics of reproduction involving economic, social, and institutional feedbacks. Fazey et al. (2011) thus emphasize that resilience does not depend solely on material endowments, but also on relational interactions and governance mechanisms, which often remain invisible in strictly quantitative approaches. In this context, linear responses based on aggregated indicators prove insufficient to grasp the complexity of adaptation, vulnerability, or maladaptation trajectories. The initial quantitative foundation of IRLA, based on indicators and composite indices, therefore constitutes a necessary but insufficient tool. Accordingly, IRLA integrates a qualitative and participatory component aimed at analyzing the feedbacks and dynamic mechanisms that structure socio-ecological system trajectories. Building on index-based results, components identified as critical are explored in greater depth with stakeholders through workshops dedicated to the construction of causal loop diagrams, making underlying systemic mechanisms explicit and identifying leverage points capable of durably altering observed trajectories.

This component builds on the system dynamics approaches developed by Sterman (2001) and on participatory modeling applied to socio-ecological change trajectories (Fazey et al., 2011). Socio-ecological systems are here understood as non-

linear, structured by multiple feedbacks, and prone to crossing critical thresholds. Stakeholder participation is a central principle of this approach, as it enables the integration of situated and distributed knowledge, reveals causal relationships absent from quantitative datasets, and produces socially robust models that are directly actionable. Causal loop diagrams, classic tools of system dynamics, are used to represent the internal structure of systems. The conventions formalized by Sterman (2001) are applied, with each arrow representing a directed and signed causal relationship, positive or negative, capable of forming reinforcing or balancing loops. CLDs thus constitute spaces of collective learning, allowing exploration of the mechanisms leading to resilience or tipping trajectories without any claim to deterministic prediction (Sterman, 2001). As such, causal loop diagrams are powerful tools for mapping and visualizing interactions among system components, highlighting interconnections, feedbacks, and the dynamics governing system evolution (Coletta et al., 2024). By making explicit vicious circles, multiplier effects, and critical thresholds, CLDs provide a shared representation of systemic dynamics and facilitate analysis of the potential effects of proposed actions. They therefore constitute a relevant support for strategic decision-making and territorial planning (Coletta et al., 2024). Causal analysis also makes it possible to assess the relative predominance of stabilizing versus degrading feedback loops, a crucial element for intervention prioritization (Fazey et al., 2011). This dynamic reading complements the quantitative assessment derived from the composite index by explaining the persistence of certain vulnerabilities, locating tipping-point risks, and clarifying the conditions for transitions toward adaptive trajectories.

One of IRLA's methodological specificities lies in the close articulation between participatory causal analysis and the spatial and statistical diagnosis derived from the composite index, with causal loop diagrams serving to explain the patterns revealed by indicator-based analysis. The approach relies on a progressive participatory process, beginning with the identification of observable symptoms of the socio-ecological system, such as declining agricultural yields, from which underlying dynamics are explored. Starting from a simple question posed to the group, for example, "why are harvests declining?", participants collectively elicit perceived causes, such as erosion, lack of inputs, deforestation, or land pressure. These variables are then connected through causal relationships within CLDs, with each link characterized by a positive or negative polarity in order to make visible the chains of causation and feedbacks governing system evolution. The analysis then seeks to identify leverage points (Meadows, 1999), defined here as upstream variables involved in multiple feedback loops. These strategic nodes make it possible to act on the structural causes of observed dynamics rather than on their superficial manifestations. Acting on such leverage points generates multiplier effects at the system scale, unlike interventions focused solely on symptoms. For instance, isolated reforestation actions that ignore poverty or forage shortages fail to durably reduce deforestation. The challenge is therefore not to break a single loop through a one-off solution, but to transform system trajectories by combining coherent and mutually reinforcing interventions. Deforestation illustrates these systemic dynamics, resulting from land pressure, survival strategies, lack of economic alternatives, and fragile governance, which renders isolated actions such as reforestation insufficient. The objective is to convert negative reinforcing loops into virtuous dynamics by combining complementary interventions, such as developing alternatives to charcoal production, improving forage systems, securing land tenure, and strengthening community governance. Resilience thus rests on the coherent articulation of economic, social, ecological, and institutional levers. Any intervention must be assessed to avoid unintended effects that may displace vulnerabilities rather than reduce them, as isolated actions can generate new tensions. From this perspective, causal loop diagrams serve a dual function: understanding the self-reinforcing mechanisms of vulnerability and identifying the levers capable of reversing them.

Each leverage point is translated into an action sheet specifying its objective, feasibility, involved actors, associated risks, and expected resilience gains. Periodic monitoring makes it possible to assess whether implemented actions initiate improvement trajectories or generate new negative feedbacks. As illustrated by Wu (2021), resilience planning can either reinforce the existing state by improving absorption and response capacities, or lead to the emergence of a new equilibrium better

adapted to future risks once stability thresholds are exceeded.

5. Discussion and conclusion

IRLA builds on the body of empirical work on resilience based on so-called objective approaches, in the sense defined by Jones (2019), by assessing resilience through indirect metrics and socio-economic proxies (Sharifi, 2016). These inputs, derived from surveys, geospatial data, and participatory contributions, make it possible to integrate dimensions that remain weakly operationalized in existing frameworks, notably psycho-cultural dimensions such as local perceptions, collective learning, trust, and social cohesion (Tariq et al., 2021). The aggregation of these data into a composite structural index, combined with a participatory analysis of socio-ecological dynamics, positions IRLA as a hybrid tool capable of capturing attenuation trajectories and regulation mechanisms that are generally difficult to measure (Djament-Tran et al., 2011), while seeking to articulate and operationalize well-established theoretical frameworks. The decision to prioritize, in this initial phase, structural and comparable measures responds to objectives of robustness, readability, and operational relevance in contexts characterized by high uncertainty, institutional instability, and limited data availability. This approach makes it possible to produce a coherent and reproducible diagnosis of latent resilience capacities at a given point in time, providing a solid basis for ex ante decision support and territorial prioritization (Annex 5).

IRLA is thus conceived as a strategic framing tool intended to guide public policies and investments, rather than as a predictive instrument. Landscape-level scores enable the identification of territorial syndromes of low resilience and support the prioritization of interventions. In this respect, IRLA is particularly well suited to large-scale mapping and inter-territorial comparison of socio-ecological landscapes. Its use is conceived in a complementary logic with qualitative, participatory, and prospective approaches, which enrich the interpretation of results. An initial data collection phase made it possible to produce a baseline of landscape resilience for a pilot territory, providing a first proof of concept. However, this phase remains exploratory and is primarily intended to lay the foundations for a structured monitoring framework. In this regard, several avenues for methodological enhancement have been identified, without calling into question the conceptual core of the index, in order to support the future evolution of IRLA.

Limitations. Beyond its exploratory status, the approach presents several limitations that should be made explicit. First, as with any composite index, results depend on normative choices regarding indicator selection, normalization, weighting, and aggregation; the robustness of landscape rankings to these choices has not yet been tested and should be assessed through sensitivity and uncertainty analysis, for instance by comparing quantile-based and min–max normalization, equal versus expert weights, and arithmetic versus geometric aggregation, following established guidance on composite indicators (OECD & JRC, 2008; Saisana et al., 2005). Second, the weighted arithmetic mean is fully compensatory and may mask critical deficits in a single dimension. Third, discretization into eleven ordinal classes trades measurement precision for comparability and interpretability, and quantile-based classes are relative to the reference sample rather than absolute. Fourth, several dimensions rely on declarative data that remain exposed to social desirability and recall biases despite the use of behavioral items. Finally, the index is cross-sectional and does not, in its current form, measure realized resilience after a shock.

- In line with the typology proposed by Winderl (2014), IRLA is positioned at Phase 4, as an operational index applied to a limited area. Progression to Phase 5 would require integration into national monitoring routines, while reaching Phase 6 would presuppose empirical validation against real shocks and the demonstration of predictive capacity. Yet resilience indices are still rarely confronted with observed post-disturbance trajectories, which limits the assessment of their ability to capture actual resilience mechanisms (Reghezza-Zitt et al., 2012). This limitation relates to the issue of circularity inherent in composite indices, whereby resilience is defined through the very variables assumed to produce it, such that score improvements tend to reflect changes in measured dimensions rather than genuine increases in resilience (Cumming, 2011). Although some approaches, such as the

Community Disaster Resilience Index, have sought to reduce this bias through empirical validation based on observed post-disaster losses (Yoon et al., 2016), such efforts remain rare and methodologically demanding.

- The integration of complementary modules addressing risk perceptions, learning capacities, and trust dynamics would make it possible to refine understanding of adaptation and transformation mechanisms in socio-ecological systems.
- Linking the index with participatory scenarios and, ultimately, with dynamic models opens promising perspectives for analyzing not only levels of resilience but also potential trajectories and the conditions for transitions toward more desirable states.
- An iterative process for improving methodological choices related to indicator selection, weighting, and aggregation, based on inter-territorial comparisons and feedback from implementation experience.
- Finally, research on participatory modeling shows that the simple transfer of tools to external managers often results in limited appropriation and a technocratic perception of the approach (Chlous-Ducharme & Gourmelon, 2011). This argues for continuous support mechanisms based on the co-interpretation of results, the facilitation of prospective workshops, and tailored training cycles. In a context marked by the proliferation of more than 271 resilience tools identified in 2021 (Tan, 2021), most of which lack empirical validation, the future improvement of IRLA depends less on adding indicators than on consolidating spaces for dialogue that foster shared understanding and collective appropriation.

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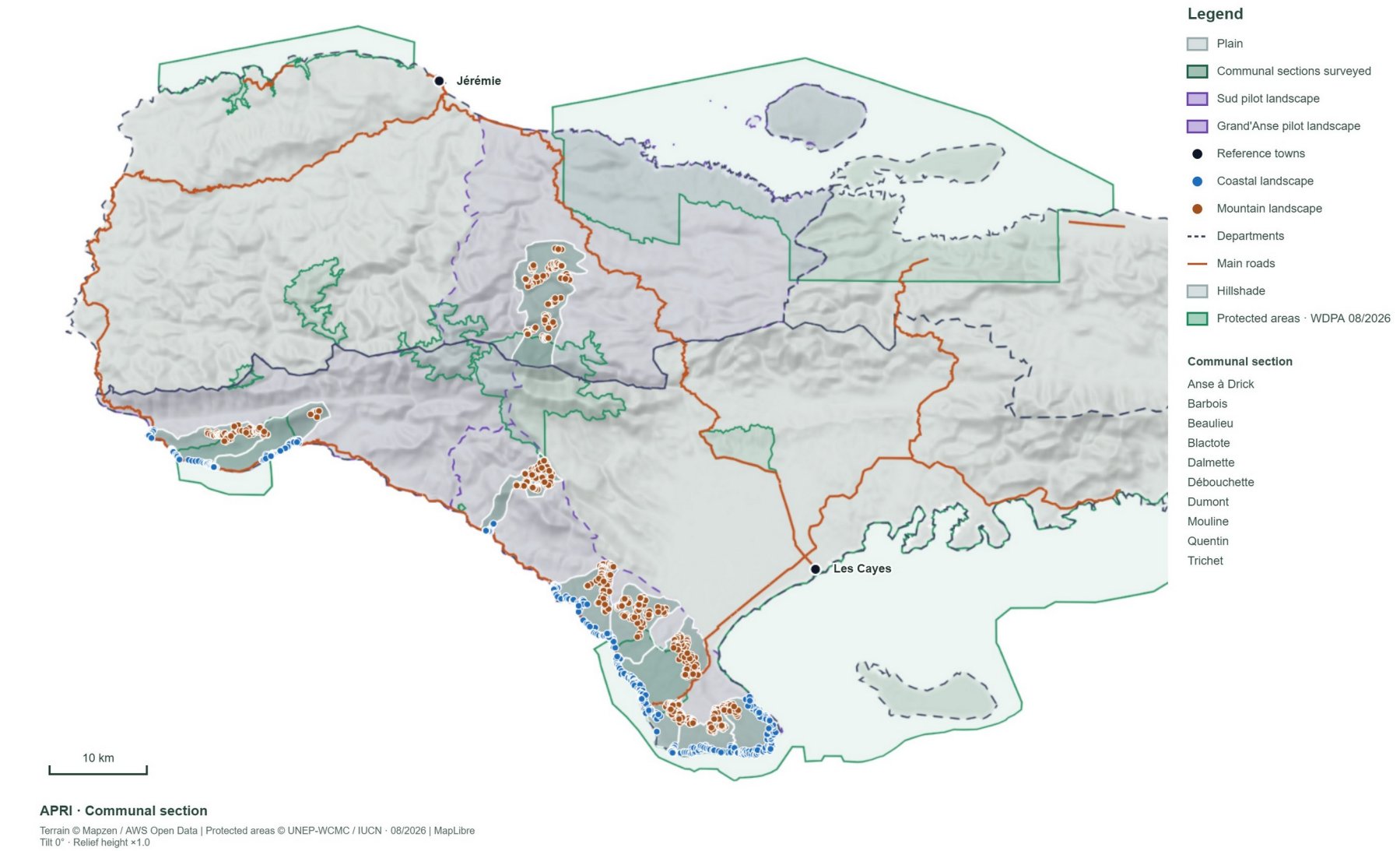
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Annex 1. Location of the study area: pilot landscapes of the Greater South



Source: Author, using open-source software (QGIS).

Annex 2. Dimensions, indicators, metrics, scoring scales and weights of the IRLA index

CE: comparative-empirical benchmark (quantile-based classes, reference sample); LN: locally normative benchmark; CI: composite item score. Weights are the relative weights W_i defined in Equation (6).

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
I. PHYSICAL AND INFRASTRUCTURAL DIMENSION						
Safely Managed Sanitation Services (SDG 6.2.1.a)	Capacity and resources	Absorption	Households (%) using improved and non-shared sanitation facilities (flush toilets connected to sewer or septic systems, ventilated improved pit latrines, pit latrines with slab, and composting toilets). Open defecation, pit latrines without slab, buckets and open drains are classified as non-improved.	Reduction of waterborne diseases, lower health expenditures, productivity preservation, environmental contamination control, dignity and well-being.	**CE** : 0 (≤ 15,8 %), 1 (15,8–26,0 %), 2 (26,0–32,0 %), 3 (32,0–45,0 %), 4 (45,0–53,0 %), 5 (53,0–63,0 %), 6 (63,0–74,0 %), 7 (74,0–84,0 %), 8 (84,0–92,0 %), 9 (92,0–97,0 %), 10 (≥ 97,0 %)	3.42
Access to Improved Drinking Water (SDG 6.1.1)	Capacity and resources	Absorption	Households (%) using improved water sources (piped water, boreholes, protected wells, protected springs, rainwater, kiosks and packaged water) and requiring ≤30 minutes round trip to collect drinking water. Non-improved sources include unprotected wells, unprotected springs and surface water.	Disease prevention, time savings, drought and cyclone coping, household stability, reduced conflict over resources.	**CE :** 0(≤18,9%), 1(18,9–31,9%), 2(31,9–47,9%), 3(47,9–64,3%), 4(64,3–74,0%), 5(74,0–81,8%), 6(81,8–92,0%), 7(92,0–96,7%), 8(96,7–98,9%), 9(98,9–99,8%), 10(≥99,8%)	4.62
Access to Electricity (SDG 7.1.1)	Capacity and resources	Absorption	Households (%) whose main source of lighting is public grid electricity or solar systems. Alternative or temporary sources such as rechargeable bulbs, kerosene lamps, candles and generators are excluded.	Food preservation, emergency lighting, operation of health services, water pumping, communication continuity….	**CE :** 0(≤27,4%), 1(27,4–43,6%), 2(43,6–60,2%), 3(60,2–80,1%), 4(80,1–82,4%), 5(82,4–85,1%), 6(85,1–87,7%), 7(87,7–90,9%), 8(90,9–94,1%), 9(94,1–96,9%), 10(≥96,9%)	3.06
Access to Clean Fuels and Technologies for Cooking (SDG 7.1.2)	Capacity and resources	Absorption	Households (%) whose primary cooking fuel or technology is classified as clean (gas and solar cookers). Solid fuels (firewood and charcoal) are classified as non-clean.	Reduced indoor air pollution, health risk reduction, deforestation pressure reduction, fuel cost stability, household labor savings.	**CE :** 0(0–5%), 1(5,1–14,5%), 2(16,4–42,7%), 3(44,1–64,8%), 4(66,4–86,8%), 5(87,9–94,2%), 6(94,4–99,5%), 7(99,7–100%), 8(100%), 9(100%), 10(100%)	1.82
Population Covered by a Mobile Network (SDG 9.c.1)	Capacity and resources	Absorption ; Anticipation	Households (%) reporting access to a mobile signal at home (at least 2G), including intermittent access. Coverage is measured independently of service quality or user satisfaction.	Information access, early warning reception, crisis communication, coordination of emergency response, access to digital services.	**CE :** 0(0–87.0%), 1(87.52–92.0%), 2(92.5–96.5%), 3(96.8–98.0%), 4(98.0–99%), 5(99–99%), 6(99–99.7%), 7(99.75–99.9%), 8(99.9–100%), 9(100%), 10(100%)	1.82
Population Living Within 2 km of an All-Season Road (SDG 9.1.1)	Capacity and resources	Absorption ; Transformation	Population (%) residing within 2 km of a road passable year-round. The indicator is derived from a combination of satellite-derived geospatial data and field-based surveys.	Emergency access, service connectivity, market integration, territorial integration, isolation reduction.	**CE :** 0(0–19.4%), 1(19.4–27.4%), 2(27.4–35.4%), 3(35.4–43.4%), 4(43.4–51.4%), 5(51.4–59.4%), 6(59.4–67.4%), 7(67.4–75.4%), 8(75.4–83.4%), 9(83.4–91.4%), 10(91.4–100%)	3.24
Housing Overcrowding Rate (SDG 11.1.1)	structural fragility or system failure outcomes	Absorption	Households (%) living in overcrowded dwellings, calculated as the share of households with a persons-per-room ratio strictly greater than 3.	Disease transmission risk, evacuation difficulty, fire hazard exposure, limited housing adaptability, psychosocial stress.	**CE** : 0(≥54%), 1(44–53%), 2(39–43%), 3(34–38%), 4(28–33%), 5(23–27%), 6(17–22%), 7(9–16%), 8(5–8%), 9(3–4%), 10(1–2%)	2.10
Housing Structural Quality Index (Walls and Roofing Materials)	Capacity and resources	Absorption	Composite indicator of dwelling structural robustness based on wall and roofing materials. 1 for stone or blocks, 0.5 for earth or planks, and 0 for straw, wattle, tarpaulins or no dwelling. 1 for concrete slabs, 0.75 for metal sheets, 0.25 for vetiver, and 0 for plastic.	Structural safety, hazard resistance, wind and rainfall protection, seismic vulnerability reduction, shelter durability.	**LN** : 0(≤5%), 1(5–15%), 2(15–25%), 3(25–35%), 4(35–45%), 5(45–55%), 6(55–65%), 7(65–75%), 8(75–85%), 9(85–95%), 10(≥95%)	2.89
Availability of School Canteen Services	Capacity and resources	Absorption	Primary schools (%) equipped with a permanent school feeding program	Child nutrition security, school attendance incentives, learning, household food burden reduction.		2.89

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Access to Drinking Water in Schools	Capacity and resources	Absorption	Primary and secondary schools (%) with an improved drinking water source	Student health protection, hygiene practices, absenteeism reduction, education continuity.		3.80
Adequate Sanitation Facilities in Schools	Capacity and resources	Absorption	Primary and secondary schools (%) equipped with adequate sanitation facilities including a water point for handwashing	Infection prevention, hand hygiene, epidemic control, gender-sensitive schooling conditions.		3.24
Availability of Operational Emergency Shelters	Capacity and resources	Absorption ; Anticipation	Communities (%) reporting the existence of at least one operational emergency shelter	Population protection capacity, evacuation readiness, disaster response infrastructure, mortality reduction.		2.40
Access to a Health Facility Within 30 Minutes	Capacity and resources	Absorption	Households (%) reporting a one-way walking time of 30 minutes or less to reach the closest health infrastructure.	Early medical intervention, emergency care access, morbidity reduction, mortality prevention, epidemic containment.		3.61
Access to a Primary School Within 30 Minutes	Capacity and resources	Transformation	Households (%) reporting a one-way walking time of 30 minutes or less to the primary school attended by their children	School attendance continuity, dropout reduction, gender equity, long-term livelihood diversification.		3.42
Solid Waste Collection and Management (SDG 11.6.1)	Capacity and resources	Absorption ; Anticipation	Households (%) reporting regular collection or controlled disposal of solid waste (incineration, burial, and dumping into the sea or on vacant land are considered non-adequate practices).	Water contamination reduction, drainage protection, flood risk reduction, air quality improvement, public health protection.		2.40
Catastrophic Events Impact Index	Structural fragility or system failure outcomes	Absorption	Measures how strongly extreme events translate into human losses (deaths or affected persons per 100,000).	Human exposure, disaster severity, mortality burden, preparedness effectiveness.	**LN :** 0 (≥100), 1 (90–99), 2 (80–89), 3 (70–79), 4 (60–69), 5 (50–59), 6 (40–49), 7 (30–39), 8 (20–29), 9 (10–19), 10 (0–9)	3,05
II. INSTITUTIONAL, TECHNOLOGICAL, AND GOVERNANCE DIMENSION						
Electoral Participation	Adaptive social dynamics	Transformation	Individuals (%) who report having voted in the most recent elections and who also declare their intention to vote in the next elections	Civic engagement, institutional legitimacy, trust in governance, collective decision-making, social cohesion.	**CE :** 0 (0–7,4 %), 1 (7,4–12,4 %), 2 (12,4–15,4 %), 3 (15,4–38,4 %), 4 (38,4–51,9 %), 5 (51,9–65,4 %), 6 (65,4–72 %), 7 (72–76,3 %), 8 (76,3–81,3 %), 9 (81,3–90,8 %), 10 (90,8–100%)	*1.82*
Exposure to Public Sector Corruption	Structural fragility or system failure outcomes	Transformation	Individuals (%) who have paid a bribe to a public official during the past 12 months in order to receive a service or avoid a sanction	Service access distortion, inequality reinforcement, erosion of trust, weakened accountability, reduced collective cooperation..	**CE** : 0(≥32.0%), 1(27.6–31.0%), 2(22.8–27.0%), 3(19.0–22.0%), 4(15.5–18.0%), 5(13.0–15.0%), 6(9.5–12.6%), 7(6.1–9.2%), 8(3.7–6.0%), 9(2.2–3.6%), 10(≤2.0%)	*2.62*
Access to Early Warning Messages	Capacity and resources	Anticipation	Percentage of households reporting receipt of at least one alert message during the most recent cyclone or hurricane (SMS or phone call, radio, loudspeaker, church or market announcement, door-to-door)	Risk information access, early warning dissemination, rapid mobilization, emergency awareness...	**LN** : 0(≤5%), 1(5–15%), 2(15–25%), 3(25–35%), 4(35–45%), 5(45–55%), 6(55–65%), 7(65–75%), 8(75–85%), 9(85–95%), 10(≥95%)	*4.00*
Disaster Preparedness Participation	Capacity and resources	Anticipation	Households (%) participation in disaster preparedness activities, including drills or awareness meetings on shelters, contingency stocks and document protection, over the past 12 months.	Risk awareness, practical readiness, household planning, community coordination, emergency response skills.		*2.68*
Civil Birth Registration	Capacity and resources	Transformation	Children (%) born in the communal section who possess an official birth certificate	Legal identity, access to rights and services, education enrollment, healthcare access, institutional inclusion...		*3.24*

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Community Meeting Participation	adaptive social dynamics	Transformation ;	Households (%) reporting participation in at least one community meeting during the last 12 months	Information circulation, collective deliberation, local coordination, shared decision-making.		*1.69*
Local Governance Membership	Adaptive social dynamics	Transformation	Households (%) reporting participation in a community governance body (management committee, vigilance committee, local association, etc.)	Local leadership, coordination capacity, rule enforcement, collective organization		*1.19*
Community Organization Membership	adaptive social dynamics	Transformation ; Absorption	Households (%) reporting membership in a community-based organization (local association, neighborhood committee, mutual aid group, cooperative).	Social capital, mutual aid structures, self-organization, collective mobilization, adaptive capacity		*1.44*
Local Risk Management Committees	Adaptive social dynamics	Anticipation ; Absorption	Communities (%) reporting the existence of at least one local risk management committee (COUC/COUL in Haiti)	Local coordination, preparedness planning, disaster governance, collective response organization		*2.89*
Accredited Schools	Capacity and resources	Transformation	Schools (%) holding an official state license or certification to operate	Education quality assurance, institutional presence, human capital formation, trust in public authority		*2.89*
Satisfaction with Public Services	Adaptive social dynamics	Transformation	Individuals (%) satisfied with their most recent experience with local public following services ; health, education and administrative services	Service reliability perception, institutional credibility, access equity, willingness to cooperate with authorities		*1.96*
Lack of Emergency Plans	Structural fragility or system failure outcomes	Anticipation	Communities (%) reporting no emergency or contingency plan	Lack of anticipation, uncoordinated response, delayed action, institutional fragility	**LN** : 0(≥95%), 1(85–95%), 2(75–85%), 3(65–75%), 4(55–65%), 5(45–55%), 6(35–45%), 7(25–35%), 8(15–25%), 9(5–15%), 10(≤5%)	*2.01*
III. ENVIRONMENTAL AND ECOLOGICAL DIMENSION						
Dry-Season Vegetation Stability (NDVI)	Capacity and resources	Absorption	Assesses vegetation activity during the most limiting season by comparing dry-season NDVI to a long-term baseline.	Dry-season biomass persistence, soil moisture retention, erosion control, forage availability	**LN :** 0 (≤ −45%), 1 (−45 to −40%), 2 (−40 to −35%), 3 (−35 to −30%), 4 (−30 to −25%), 5 (−25 to −20%), 6 (−20 to −15%), 7 (−15 to −10%), 8 (−10 to −5%), 9 (−5 to 0%), 10 (0 to +5%)	2.40
Dry-Season Vegetation and Soil Moisture (NDMI)	Capacity and resources	Absorption	Evaluates the capacity of vegetation and soils to retain moisture during the dry season, relative to long term NDMI baseline conditions	Plant water availability, early drought signal, hydrological buffering, soil moisture conservation		2.25
Dry-Season Surface Water Stability (NDWI)	Capacity and resources	Absorption	Measures the persistence or contraction of surface water bodies during the driest period of the year relative to long-term NDWI reference conditions.	Surface water persistence, water access continuity, livestock and irrigation support, hydrological buffering		2.25
Vegetation Health Index (VHI)	Capacity and resources	Absorption	Integrates thermal and hydric stress into a single signal of vegetation functioning	Combined heat and moisture stress, photosynthetic activity maintenance, biomass production continuity	**LN :** 0 (VHI ≤ 10), 1 (10–20), 2 (20–30), 3 (30–40), 4 (40–50), 5 (50–60), 6 (60–70), 7 (70–80), 8 (80–90), 9 (90–95), 10 (>95)	2.40
Enhanced Vegetation Index (EVI)	Capacity and resources	Absorption	Measures vegetation vigor, biomass density and photosynthetic performance under favorable canopy conditions.	Biomass productivity, photosynthetic robustness, buffering capacity	**LN :** 0 (< 0.20), 1 (0.20–0.25), 2 (0.25–0.30), 3 (0.30–0.35), 4 (0.35–0.40), 5 (0.40–0.45), 6 (0.45–0.48), 7 (0.48–0.50), 8 (0.50–0.53), 9 (0.53–0.55), 10 (> 0.55)	2.40
Soil-Adjusted Vegetation Index (SAVI)	Capacity and resources	Absorption	Measure vegetation condition in dry and semi-arid areas where much of the ground is bare soil. Less influence of exposed soil > easier to distinguish real vegetation health from soil brightness.	Soil protection in drylands, erosion resistance, drought tolerance	**LN :** 0 (< 0.15), 1 (0.15–0.18), 2 (0.18–0.22), 3 (0.22–0.26), 4 (0.26–0.30), 5 (0.30–0.34), 6 (0.34–0.38), 7 (0.38–0.41), 8 (0.41–0.44), 9 (0.44–0.45), 10 (> 0.45)	2.40

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Atmospherically Resistant Vegetation Index (VARI)	Capacity and resources	Absorption	Helps distinguish real vegetation from sand, bright soils or turbid water when other methods fail.	Coastal vegetation persistence, salinity and brightness tolerance erosion or high surface reflectance stress	**LN :** 0 (< 0.05), 1 (0.05–0.07), 2 (0.07–0.09), 3 (0.09–0.11), 4 (0.11–0.13), 5 (0.13–0.15), 6 (0.15–0.17), 7 (0.17–0.19), 8 (0.19–0.21), 9 (0.21–0.25), 10 (> 0.25)	2.40
Fractional Vegetation Cover (FVC)	Capacity and resources	Absorption	Ground surface (%) effectively covered by vegetation	Soil cover, runoff reduction, microclimate regulation erosion hydrological processes	**LN :** 0 (< 20%), 1 (20–25%), 2 (25–30%), 3 (30–35%), 4 (35–45%), 5 (45–55%), 6 (55–65%), 7 (65–75%), 8 (75–80%), 9 (80–85%), 10 (> 85%)	2.40
Seasonal land surface temperature anomaly (LST)	Structural fragility or system failure outcomes	Anticipation	Deviations of seasonal land surface temperature from long term reference conditions.	Thermal pressure, evapotranspiration demand, soil moisture depletion, crop heat stress	**LN :** 0 (≥ +15%), 1 (+13 to +15%), 2 (+11 to +13%), 3 (+9 to +11%), 4 (+7 to +9%), 5 (+5 to +7%), 6 (+3 to +5%), 7 (+1 to +3%), 8 (−1 to +1%), 9 (−3 to −1%), 10 (≤ −3%)	2.40
Temperature Condition Index (TCI)	Structural fragility or system failure outcomes	Anticipation	Current normal conditions vs historical tolerance range of ecosystems.	Heat stress exposure, vegetation functioning limitation, interaction with water stress, thermal vulnerability	**LN :** 0 (TCI ≤ 10), 1 (10–20), 2 (20–30), 3 (30–40), 4 (40–50), 5 (50–60), 6 (60–70), 7 (70–80), 8 (80–90), 9 (90–95), 10 (>95)	2.40
Precipitation Condition Index (PCI)	Structural fragility or system failure outcomes	Anticipation	Current rainfall compares to historical seasonal norms.	Rainfall availability, soil moisture recharge, crop growth conditions, drought pressure modulation	**LN :** 0 (PCI ≤ 10), 1 (10–20), 2 (20–30), 3 (30–40), 4 (40–50), 5 (50–60), 6 (60–70), 7 (70–80), 8 (80–90), 9 (90–95), 10 (>95)	2.40
Anomalous Aridity Index	Structural fragility or system failure outcomes	Anticipation	Imbalance between dry and wet conditions relative to a long term equilibrium.	Hydroclimatic imbalance, drought or excess moisture exposure, climate anomaly sensitivity	**LN :** 0 (AAI ≤ −0.80), 1 (−0.80 to −0.65), 2 (−0.65 to −0.50), 3 (−0.50 to −0.35), 4 (−0.35 to −0.20), 5 (−0.20 to 0.00), 6 (0.00 to +0.20), 7 (+0.20 to +0.35), 8 (+0.35 to +0.50), 9 (+0.50 to +0.65), 10 (≥ +0.65)	2.40
Ratio to normal precipitation	Structural fragility or system failure outcomes	Anticipation	Observed rainfall as a percentage of normal conditions.	Rainfall deficit or surplus, water availability, flood or drought exposure, production risk	**LN :** 0 <5 ou >125, 1 ≤10 → 122.5, 2 ≤20 → 120, 3 ≤30 → 117.5, 4 ≤40 → 115, 5 ≤50 → 112.5, 6 ≤60 ou ≥110, 7 ≤70 → 107.5, 8 ≤80 → 105, 9 ≤90 → 102.5, 10 95–100	2.40
Standardized Precipitation Index (SPI)	Structural fragility or system failure outcomes	Anticipation	Rarity and severity of precipitation anomalies relative to a historical distribution.	Meteorological drought intensity, persistence of rainfall deficits, hydroclimatic stress duration	**LN :** 0 (SPI ≤ −2.0), 1 (−2.0 < SPI ≤ −1.9), 2 (−1.9 < SPI ≤ −1.8), 3 (−1.8 < SPI ≤ −1.7), 4 (−1.7 < SPI ≤ −1.6), 5 (−1.6 < SPI ≤ −1.5), 6 (−1.5 < SPI ≤ −1.4), 7 (−1.4 < SPI ≤ −1.3), 8 (−1.3 < SPI ≤ −1.2), 9 (−1.2 < SPI ≤ −1.1), 10 (−1.1 < SPI ≤ −1.0)	2.40
Species richness (Od, Bi, Po)	Capacity and resources	Absorption	Number of distinct species observed within each taxonomic group, namely odonates (Od), birds (Bi) and pollinators (Po), over a standardized sampling effort.	Ecosystem integrity, functional redundancy, habitat connectivity, recovery capacity, ecosystem service continuity	**LN :** 0 Absence or a single highly tolerant species, indicating severely degraded conditions across aquatic habitats, landscapes, or floral resources ; 10 High species richness with many sensitive and specialist species, indicating near reference ecological conditions and strong recovery potential across habitats	3.42

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Simpson diversity index (1–D)	Capacity and resources	Absorption	How evenly individuals are distributed among species in a community. It captures dominance, not just richness. High values mean no single species dominates.	Dominance reduction, functional balance, disturbance buffering, ecosystem process stability....	**LN** : (Od) 0 (0–.05), 1 (.06–.1), 2 (.11–.18), 3 (.19–.3), 4 (.31–.45), 5 (.46–.6), 6 (.61–.7), 7 (.71–.79), 8 (.8–.85), 9 (.86–.89), 10 (≥ .9) ; (Bi) 0 (0–.1), 1 (.11–.18), 2 (.19–.26), 3 (.27–.38), 4 (.39–.52), 5 (.53–.65), 6 (.66–.75), 7 (.76–.87), 8 (.88–.92), 9 (.93–.96), 10 (≥ .97) ; (Po) 0 (0–.05), 1 (.06–.1), 2 (.11–.18), 3 (.19–.3), 4 (.31–.45), 5 (.46–.6), 6 (.61–.7), 7 (.71–.84), 8 (.85–.9), 9 (.91–.94), 10 (≥ .95)	2.43
Shannon diversity index (H')	Capacity and resources	Absorption	Measures both species richness and evenness. How many species are present and how evenly individuals are distributed among them ($H' = -\Sigma (p_i \times \ln(p_i))$	Ecological complexity, evenness of functions, adaptive capacity, process robustness...	LN : (Od) 0 (0–.1), 1 (.11–.3), 2 (.31–.6), 3 (.61–.9), 4 (.91–1.2), 5 (1.21–1.5), 6 (1.51–1.8), 7 (1.81–1.99), 8 (2–2.2), 9 (2.21–2.39), 10 (≥ 2.4) (Bi) 0 (0–.15), 1 (.16–.4), 2 (.41–.8), 3 (.81–1.2), 4 (1.21–1.6), 5 (1.61–2), 6 (2.01–2.4), 7 (2.41–2.79), 8 (2.8–3), 9 (3.01–3.29), 10 (≥ 3.3) (Po) 0 (0–.1), 1 (.11–.3), 2 (.31–.6), 3 (.61–.9), 4 (.91–1.2), 5 (1.21–1.5), 6 (1.51–1.9), 7 (1.91–2.29), 8 (2.3–2.59), 9 (2.6–2.79), 10 (≥ 2.8)	2.62
Phylogenetic diversity	Capacity and resources	Absorption	Evolutionary breadth and response diversity.	Evolutionary response diversity, functional insurance effect, long-term adaptability, disturbance resistance	**LN** : 0 (Od) Single family present, species from closely related lineages, (Bi) One or two dominant families of closely related generalist species, (Po) One dominant taxonomic group with no phylogenetically distinct lineages, 10 (Od) Multiple families spanning distant evolutionary lineages, (Bi) Multiple avian families and orders reflecting deep evolutionary divergence and high phylogenetic representation	2.25
Relative abundance	Capacity and resources	Absorption	Proportion that each species or functional group represents within the total number of individuals observed in a community.	Intensity of ecological processes, pollination and regulation capacity, recovery potential...	**LN :** (Od) 0 absence or ≤2 tolerant individuals, near-null activity and degraded habitat, 10 >50 individuals per transect with sensitive families; (Bi) 0 absence or a single tolerant generalist, no insectivores or nectarivores, 10 >30 species with high abundance including sensitive and endemic birds; (Po) 0 absence or a few opportunistic individuals, no wild bees or butterflies, no floral activity, 10 >100 individuals with high diversity of bees, butterflies and sensitive nocturnal sphinx	2.17
Functional diversity	Capacity and resources	Absorption	Diversity of ecological traits and roles represented in a community.	Trait complementarity, response diversity, disturbance buffering, ecosystem process continuity...	**LN :** 0 (Od) Single tolerant function, (Bi) One generalist guild only, no forest insectivores, frugivores or nectarivores, (Po) Single opportunistic functional group; 10 (Od) Full functional spectrum, diverse flight and hunting strategies, (Bi) Complete and balanced guilds with functional trophic network, (Po) Broad functional diversity with specialist bees, butterflies and sensitive nocturnal sphinx moths	1.71

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Crop Diversity Index (CDI)	Capacity and resources	Absorption Anticipation	How diversified household production systems are. CDI = 1 − (p_1^2 + p_2^2 + … + p_n^2)	Production diversification, risk spreading, food availability stability, income smoothing, biological shock buffering	**LN :** 0 = 0, 1 > 0–0.40, 2 > 0.40–0.50, 3 > 0.50–0.60, 4 > 0.60–0.65, 5 > 0.65–0.70, 6 > 0.70–0.75, 7 > 0.75–0.80, 8 > 0.80–0.85, 9 > 0.85–0.90, 10 > 0.90–1.00	2.62
Forest Cover Change Rate	Structural fragility or system failure outcomes	Absorption Anticipation	Forest cover shrinking or expanding / current pace of change worse or better than the historical norm.	Soil protection, hydrological regulation, carbon storage continuity, habitat availability….	**LN :** 0 (< −3), 1 (−3 to −2), 2 (−2 to −1.5), 3 (−1.5 to −1), 4 (−1 to −0.75), 5 (−0.75 to −0.5), 6 (−0.5 to −0.25), 7 (−0.25 to 0), 8 (0 to 0.25), 9 (0.25 to 0.5), 10 (≥ 0.5)	2.33
Marine–Coastal PA Coverage	Capacity and resources	Absorption Transformation	Coastal and marine ecosystems (%) that are legally designated and managed for conservation	Stock recovery space, habitat safeguarding, coastal buffering, food supply stability	**LN** (%) : 0 (0), 1 (1–3), 2 (4–6), 3 (7–9), 4 (10–12), 5 (13–15), 6 (16–18), 7 (19–21), 8 (22–24), 9 (25–27), 10 (28 and above)	2.10
Terrestrial–Freshwater PA Coverage	Capacity and resources	Absorption Transformation	Land and inland water ecosystems (%) that are under long term legal protection.	Ecosystem representativity, flood and drought mitigation, service continuity		2.01
Mangrove surface	Capacity and resources	Absorption	Spatial extent of mangrove expressed as a (%) of historical surface.	Wave attenuation, storm surge buffering, fisheries nursery support	**LN :** 0 (< 20%), 1 (20–30%), 2 (30–40%), 3 (40–50%), 4 (50–60%), 5 (60–70%), 6 (70–75%), 7 (75–80%), 8 (80–85%), 9 (85–90%), 10 (> 90%)	2.25
Seagrass surface	Capacity and resources	Absorption	Spatial extent of seagrass meadows expressed as a (%) of historical surface.	Seabed stabilization, water clarity, juvenile habitat, coastal disturbances, fisheries productivity		2.33
Soil Erosion and Loss	Structural fragility or system failure outcomes	Absorption	Rate at which landscapes lose productive soil using spatially explicit erosion models, typically based on the Universal Soil Loss	Soil fertility loss, runoff regulation, sediment delivery, agricultural productivity risk	**LN (**t ha yr), 0 (200 and above), 1 (150–199), 2 (120–149), 3 (90–119), 4 (60–89), 5 (40–59), 6 (25–39), 7 (15–24), 8 (8–14), 9 (3–7), 10 (0–2)	3.42
Degraded Land (SDG 15.3.1)	Structural fragility or system failure outcomes	Absorption	Land (%) affected by degradation processes (vegetation condition, soil erosion risk, and productivity)	Long-term productivity decline, soil instability, drought and flood sensitivity	**LN:** 0 (≥50%), 1 (45–49%), 2 (40–44%), 3 (35–39%), 4 (30–34%), 5 (25–29%), 6 (20–24%), 7 (15–19%), 8 (10–14%), 9 (5–9%), 10 (0–4%)	3.42
Chlorophyll-a concentration	Structural fragility or system failure outcomes	Absorption	Phytoplankton biomass as a proxy for nutrient loading and trophic imbalance in aquatic systems.	Nutrient loading pressure, eutrophication risk, fisheries productivity constraint	**LN :** Clorophyll-a concentration, mg m^{-3}, 0 (> 20), 1 (15–20), 2 (12–15), 3 (10–12), 4 (8–10), 5 (6–8), 6 (4–6), 7 (3–4), 8 (2–3), 9 (1–2), 10 (< 1)	2.25
Marine macroplastic	Structural fragility or system failure outcomes	Absorption	Prevalence of visible macroplastic	Failures in waste management, wildlife mortality, health risks and impacts on tourism and coastal livelihoods.	**LN (%) :** 0 (> 40), 1 (35–40), 2 (30–35), 3 (25–30), 4 (20–25), 5 (15–20), 6 (10–15), 7 (5–10), 8 (2–5), 9 (1–2), 10 (< 1)	2.25
Water turbidity (NDTI)	Structural fragility or system failure outcomes	Absorption	Sediment loads and water clarity	Sediment regulation, upstream erosion, runoff control, aquatic habitat quality, water use continuity	**LN :** 0 (−1 to −0.8), 1 (−0.8 to −0.6), 2 (−0.6 to −0.4), 3 (−0.4 to −0.2), 4 (−0.2 to 0), 5 (0 to 0.1), 6 (0.1 to 0.2), 7 (0.2 to 0.35), 8 (0.35 to 0.5), 9 (0.5 to 0.7), 10 (>0.7)	1.78
Functional Connectivity (PC(d*))	Capacity and resources	Absorption	Probability that habitat patches are functionally connected based on species dispersal capacity. PC(d*) represents connectivity at a biologically realistic dispersal distance for target species. PC(d*/2)	Dispersal success, metapopulation persistence, recolonization probability	**LN :** 0 (PC < .1), 1 (.1 to .15), 2 (.15 to .2), 3 (.2 to .25), 4 (.25 to .3), 5 (.3 to .35), 6 (.35 to .4), 7 (.4 to .45), 8 (.45 to .5), 9 (.5 to .6), 10 (> .6)	2.40

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Low-Mobility Connectivity (PC(d*/2))	Capacity and resources	Absorption	captures connectivity under more restrictive movement conditions for low-mobility species.	Sensitive species isolation risk, short-range movement limitation	**LN :** 0 (PC < .05), 1 (.05 to .1), 2 (.1 to .15), 3 (.15 to .2), 4 (.2 to .25), 5 (.25 to .3), 6 (.3 to .35), 7 (.35 to .4), 8 (.4 to .45), 9 (.45 to .55), 10 (> .55)	2.40
Structural Connectivity (IIC)	Capacity and resources	Absorption	Structural connectivity of habitat patches calculated using graph-based analysis, where patches are nodes and links represent potential connections constrained by dispersal distance. IIC(d) reflects connectivity within a realistic dispersal threshold	Species movement, recolonization capacity, genetic exchange, fragmentation limitation	**LN :** 0 (≤ .05), 1 (.06, .15), 2 (.16, .25), 3 (.26, .35), 4 (.36, .45), 5 (.46, .55), 6 (.56, .65), 7 (.66, .75), 8 (.76, .85), 9 (.86, .95), 10 (≥ .96)	2.17
Constrained Structural Connectivity (IIC(d*))	Capacity and resources	Absorption		Patch spacing, fragmentation constraint, spatial continuity		2.17
Edge density	Structural fragility or system failure outcomes	Absorption	Intensity of habitat fragmentation by quantifying how much habitat edge exists relative to landscape area	Edge effects exposure, microclimatic stress, invasive pressure	**LN :** 0 (> 100 m/ha), 1 (90–100 m/ha), 2 (80–90 m/ha), 3 (70–80 m/ha), 4 (60–70 m/ha), 5 (50–60 m/ha), 6 (40–50 m/ha), 7 (30–40 m/ha), 8 (20–30 m/ha), 9 (15–20 m/ha), 10 (< 15 m/ha)	1.86
CORE + MOVE to favorable habitat ratio	Capacity and resources	Absorption	Measures how favorable habitat is functionally allocated between stable core areas that support reproduction and persistence, and movement areas that act as corridors or stepping stones. Core areas are identified by removing edge-affected zones using a fixed buffer distance	Landscape permeability, population persistence, dispersal pathways	**LN :** 0 (< 0.30), 1 (0.30–0.35), 2 (0.35–0.40), 3 (0.40–0.45), 4 (0.45–0.50), 5 (0.50–0.55), 6 (0.55–0.60), 7 (0.60–0.65), 8 (0.65–0.70), 9 (0.70–0.75), 10 (> 0.75)	2.17
Total CORE area	Capacity and resources	Absorption	Total surface of high-quality interior habitat that is sufficiently large and isolated from edge effects to anchor population dynamics	Source population anchoring, long-term habitat stability	**LN :** 0 (< 5%), 1 (5–7%), 2 (7–10%), 3 (10–12%), 4 (12–15%), 5 (15–18%), 6 (18–20%), 7 (20–22%), 8 (22–25%), 9 (25–30%), 10 (> 30%)	2.17
IV. ECONOMIC, LIVELIHOODS, AND FOOD SECURITY DIMENSION						
Population holding an account (SDG 8.10.2)	Capacity and resources	Absorption	Adults (%) reporting ownership of an account in a formal financial institution, including banks, credit unions or deposit-taking microfinance institutions.	Liquidity management, consumption smoothing, risk management capacity, reduction of informal dependency, household shock coping mechanisms…	**CE**: 0(≤26.3%), 1(27.4–36.9%), 2(37.0–45.3%), 3(46.3–51.3%), 4(51.4–59.7%), 5(59.8–68.9%), 6(69.1–79.8%), 7(80.8–88.4%), 8(88.7–95.6%), 9(95.7–99.1%), 10(≥99.2%)	1.44
Access to financial services (SDG 8.10.1)	Capacity and resources	Absorption	Number of commercial bank branches and automated teller machines per 100,000 adults.	Physical access to finance, territorial equity, cash availability, transaction security, rural inclusion	**CE**: 0 (<1), 1 (1–3), 2 (3–6), 3 (6–9), 4 (9–12), 5 (12–15), 6 (15–18), 7 (18–21), 8 (21–24), 9 (24–27), 10 (≥27)	1.56
Land tenure security (SDG 1.4.2a)	Capacity and resources	Transformation	Households (%) reporting recognized rights over at least one agricultural plot, including formal titles, customary ownership, rental or sharecropping arrangements.	Asset security, protection against dispossession, investment incentives, long-term land use planning, livelihood continuity, intergenerational stability…	**CE:** 0(0.0–7.4%), 1(7.4–12.4%), 2(12.4–15.4%), 3(15.4–38.4%), 4(38.4–51.9%), 5(51.9–65.4%), 6(65.4–72.0%), 7(72.0–76.3%), 8(76.3–81.3%), 9(81.3–90.8%), 10(90.8–100.0%)	2.72
Informal employment (% of employed population)	Structural fragility or system failure outcomes	Absorption	Individuals (%) engaged in informal jobs without a stable written contract or access to social protection.	Income instability, absence of social protection, exposure to health shocks, lack of labor rights, vulnerability to economic downturns…	**CE (inverse):** 0(93.0–98.5%), 1(85.8–93.0%), 2(81.1–85.8%), 3(67.1–81.1%), 4(55.8–67.1%), 5(51.1–55.8%), 6(35.7–51.1%), 7(19.5–35.7%), 8(3.5–19.5%), 9(2.0–3.5%), 10(0.7–2.0%)	1.31
Employment rate	Capacity and resources	Absorption	Adults (%) employed, including salaried employment, daily wage labor and self-employment.	Income generation capacity, labor market integration, livelihood continuity, reduced dependency,	**CE:** 0(≤29.2%), 1(29.2–43.9%), 2(43.9–49.1%), 3(49.1–53.6%), 4(53.6–57.8%), 5(57.8–60.2%), 6(60.2–62.8%), 7(62.8–66.4%), 8(66.4–70.2%), 9(70.2–80.3%), 10(≥80.3%)	2.25

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Household debt ratio	Structural fragility or system failure outcomes	Absorption	Ratio between total household debt and total annual household income, including debts owed to banks, microfinance institutions, informal lenders and relatives.	Financial pressure, constrained investment capacity, vulnerability to income shocks, inequality amplification…	**LN:** 0 (more than 95), 1 (95–85), 2 (85–75), 3 (75–65), 4 (65–55), 5 (55–45), 6 (45–35), 7 (35–25), 8 (25–15), 9 (15–5), 10 (5 or less)	1.96
Population Below 50% of Median Income	Structural fragility or system failure outcomes	Absorption	Individuals (%) living with less than half of the national median income	Relative deprivation, limited access to basic services, restricted investment in health and education...	**CE:** 0 (more than 25), 1 (25–23), 2 (23–21), 3 (21–19), 4 (19–17), 5 (17–15), 6 (15–13), 7 (13–11), 8 (11–9), 9 (9–7), 10 (7 or less)	2.10
Remittances coverage	Capacity and resources	Absorption	Households (%) reporting receipt of domestic or international monetary transfers during the past 12 months.	External income inflows, consumption smoothing, risk sharing, temporary shock compensation…	**LN :** 0(0–2%), 1(2–5%), 2(5–8%), 3(8–12%), 4(12–16%), 5(16–20%), 6(20–25%), 7(25–30%), 8(30–40%), 9(40–50%), 10(>50%).	1.71
Households without an economic safety net	Structural fragility or system failure outcomes	Absorption	Households (%) reporting no savings, no mobilizable assets and no social or financial support mechanisms, excluding land ownership.	Absence of savings, lack of mobilizable assets, immediate exposure to income loss, inability to absorb shocks...	**LN (inverse)**: 0(≥95%), 1(85–95%), 2(75–85%), 3(65–75%), 4(55–65%), 5(45–55%), 6(35–45%), 7(25–35%), 8(15–25%), 9(5–15%), 10(≤5%)	2.68
Income Buffer < 1 Week	Structural fragility or system failure outcomes	Absorption	Households (%) whose savings or safety nets would cover less than one week of basic subsistence following income loss.	Acute vulnerability, absence of short-term buffers, rapid livelihood collapse under shocks		2.68
Chemical Input Use Rate	Structural fragility or system failure outcomes	Anticipation Absorption	Agricultural households (%) using at least one chemical input during the last production cycle, capturing overall dependence on external inputs (fertilizer, herbicide, insecticide, and fungicide).	Dependence on external inputs, exposure to price volatility, sensitivity to supply chain disruptions, soil degradation risk		1.30
Livestock Loss Rate (12 Months)	Structural fragility or system failure outcomes	Absorption	Total livestock (%) capital lost over the past 12 months, expressed as a (%) of the average herd size converted into standardized Animal Units (AU). Conversion coefficients are adult cattle = 1 AU, young cattle = 0.5 AU, equids = 0.8 AU, pigs = 0.2 AU, goats and sheep = 0.1 AU	Productive asset depletion, income discontinuity, reduced recovery capacity, increased sensitivity to climatic shocks	**LN (inverse)**: 0(≥50%), 1(45–50%), 2(40–45%), 3(35–40%), 4(30–35%), 5(25–30%), 6(20–25%), 7(15–20%), 8(10–15%), 9(5–10%), 10(≤5%).	2.25
Agricultural Production Losses	Structural fragility or system failure outcomes	Absorption	Measures agricultural production losses (%) during the last cycle, disaggregated by cereals, legumes, tubers, musaceae, vegetables and fruits	Livelihood vulnerability, cash income, caloric avaibility, diversity and micronutrient intake		2.72
Low Chemical Input Dependen	Capacity and resources	Anticipation	Agricultural households (%) not using any chemical inputs during the reference production cycle	Production autonomy, lower exposure to price shocks, soil biological functioning	**LN:** 0(≤ 5 %), 1(5–15 %), 2(15–25 %), 3(25–35 %), 4(35–45 %), 5(45–55 %), 6(55–65 %), 7(65–75 %), 8(75–85 %), 9(85–95 %), 10(≥ 95 %)	1.30
Fished biodiversity index	Capacity and resources	Absorption Anticipation	Number of distinct aquatic species fished per household across all fishing gears and locations during the reference period.	Livelihood diversification, flexibility of fishing strategies, reduced dependence on single species or habitat	**LN** : 0(1), 1(2), 2(3), 3(4), 4(5), 5(6), 6(7), 7(8), 8(9), 9(10), 10(≥11)	1.90
"Exclusive shelf (plateau) fishers"	Structural fragility or system failure outcomes	Anticipation	Fishers (%) operating exclusively on the coastal shelf or reef zone, with no offshore fishing activities.	Spatial rigidity, dependence on a single ecosystem, exposure to coastal degradation	**LN :** 0(≥95%), 1(85–95%), 2(75–85%), 3(65–75%), 4(55–65%), 5(45–55%), 6(35–45%), 7(25–35%), 8(15–25%), 9(5–15%), 10(≤5%)	2.25

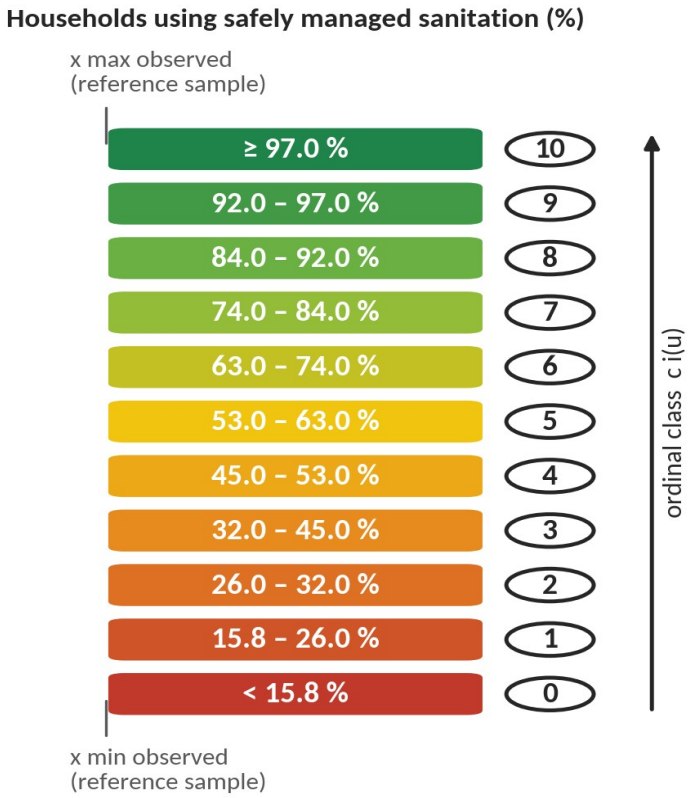
Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Destructive Fishing Practices	Structural fragility or system failure outcomes	Transformation	Fishers (%) reporting use of at least one environmentally or socially destructive fishing technique.	Stock depletion risk, habitat degradation, occupational hazards, legal exposure, long-term livelihood erosion	**LN :** 0(≤5%), 1(5–15%), 2(15–25%), 3(25–35%), 4(35–45%), 5(45–55%), 6(55–65%), 7(65–75%), 8(75–85%), 9(85–95%), 10(≥95%)	2.50
V. SOCIAL AND COMMUNITY DIMENSION						
Perception of security (SDG 16.1.4)	adaptive social dynamics	Absorption	Population (%) reporting that they feel safe walking alone at night in their community.	Mobility of people and goods, trust in local environment, access to services and markets, community confidence	**CE:** 0 (≤ 37.0%), 1 (37.1–42.9%), 2 (43.0–51.7%), 3 (51.8–61.8%), 4 (61.9–69.6%), 5 (69.7–75.0%), 6 (75.1–77.0%), 7 (77.1–82.0%), 8 (82.1–86.1%), 9 (86.2–89.5%), 10 (≥ 89.6%)	1.82
Single-parent households	Structural fragility or system failure outcomes	Absorption	Households (%) headed by a single parent.	Single-income pressure, reduced labor availability, childcare burden, limited financial flexibility	**CE:** 0 (≥ 32%), 1 (27–30%), 2 (24–27%), 3 (21–24%), 4 (18–21%), 5 (15–18%), 6 (12–15%), 7 (9–12%), 8 (6–9%), 9 (3–6%), 10 (≤ 3%)	1.86
Women's Autonomy Index	Capacity and resources	Transformation	Composite index based on a set of binary (0/1) question proxies covering income generation, ability to borrow small amounts independently, autonomy in managing everyday expenses, freedom of movement, social interactions, participation in community groups, and absence of fear when expressing disagreement.	Control over income and assets, independent decision-making, livelihood diversification, household risk management, social participation, collective action capacity	**CI:** 0, 1, 2, 3, 3.5, 4, 4.5, 5, 5.5, 6, 7	2.01
Population victim of violence	Structural fragility or system failure outcomes	Absorption	Population (%) that experienced physical, psychological or sexual violence during the last 12 months. Psychological violence includes threats, insults and intimidation, and sexual violence covers any non-consensual sexual act.	Physical harm exposure, psychological stress, disruption of daily activities, breakdown of social trust, reduced community cooperation, chronic insecurity	**LN:** 0 (≤ 5.0%), 1 (5.0–15.0%), 2 (15.0–25.0%), 3 (25.0–35.0%), 4 (35.0–45.0%), 5 (45.0–55.0%), 6 (55.0–65.0%), 7 (65.0–75.0%), 8 (75.0–85.0%), 9 (85.0–95.0%), 10 (≥ 95.0%)	2.89
Girls enrolled in school after 15	Capacity and resources	Transformation	Girls (%) aged 15 years and above enrolled in secondary or higher education.	Human capital accumulation, delayed marriage and pregnancy, access to information, economic autonomy pathways, intergenerational welfare improvement	**LN:** 0 (≤ 5.0%), 1 (5.0–15.0%), 2 (15.0–25.0%), 3 (25.0–35.0%), 4 (35.0–45.0%), 5 (45.0–55.0%), 6 (55.0–65.0%), 7 (65.0–75.0%), 8 (75.0–85.0%), 9 (85.0–95.0%), 10 (≥ 95.0%)	2.50
Bonding social capital (family and close relatives)	Adaptive social dynamics	Absorption	Households (%) that both gave and received assistance within their close family network during the last 12 months.	Immediate mutual aid, informal safety nets, internal resource redistribution, short-term shock buffering		2.35
Bridging social capital (village)	Adaptive social dynamics	Absorption, Transformation	Households (%) that both gave and received assistance within the village or extended community beyond close family during the last 12 months.	Horizontal cooperation, collective resource mobilization, information circulation		1.78
Conflicts resolved peacefully	Adaptive social dynamics	Absorption, Transformation	Community conflicts (%) settled through dialogue, mediation or intervention of recognized authorities.	Local mediation capacity, trust in institutions, avoidance of violence escalation, continuity of social relations...		2.62
Isolated population	Structural fragility or system failure outcomes	Absorption	Individuals (%) who did not participate in any collective activity or group during the last 12 months, including community work, associations, religious groups, savings groups or informal social networks.	Weak social ties, lack of mutual aid, limited information access, low participation in collective action, high vulnerability to shocks	LN: 0 (≥ 95.0%], 1 (85.0–95.0%], 2 (75.0–85.0%], 3 (65.0–75.0%], 4 (55.0–65.0%], 5 (45.0–55.0%], 6 (35.0–45.0%], 7 (25.0–35.0%], 8 (15.0–25.0%], 9 (5.0–15.0%], 10 (≤ 5.0%)	1.90

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Psychosocial isolation	Structural fragility or system failure outcomes	Absorption	Individuals (%) who report not sharing their concerns with any trusted person within the community.	Distress risk, weakened informal care networks, reduced collective coping		1.44
Teenage Motherhood	Structural fragility or system failure outcomes	Transformation	Women (%) reporting their first childbirth before the age of 19.	Interrupted education, reduced labor opportunities, constrained autonomy, long-term income limitation		2.68
VI. HUMAN DIMENSION						
Skilled Birth Attendance (SDG 3.1.2)	Capacity and resources	Absorption	Live births (%) assisted by a doctor, nurse or trained midwife during the reference period.	Maternal health protection, newborn survival, anticipatory health risk management, preservation of human capital, household demographic stability	**CE:** 0(≤40.0%), 1(40.0–46.0%), 2(46.0–52.0%), 3(52.0–58.0%), 4(58.0–64.0%), 5(64.0–70.0%), 6(70.0–76.0%), 7(76.0–82.0%), 8(82.0–88.0%), 9(88.0–94.0%), 10(≥94.0%)	2.07
Population holding a valid national ID	Capacity and resources	Transformation;	Individuals (%) aged 15 years and above possessing a valid national identification document.	Institutional inclusion, access to public services, eligibility for transfers, legal protection, mobility rights, administrative security	**CE:** 0(0.0–67.1%), 1(68.1–83.2%), 2(83.2–87.7%), 3(87.8–90.6%), 4(91.6–92.9%), 5(93.6–95.5%), 6(95.5–96.4%), 7(96.6–98.0%), 8(98.2–99.0%), 9(99.1–99.3%), 10(99.4–100.0%)	2.25
Neonatal mortality ratio (SDG 3.2.2)	Structural fragility or system failure outcomes	Absorption	Probability of death during the first 28 days of life per 1,000 live births.	Early life vulnerability, effectiveness of perinatal care, health system performance, long-term human capital loss	**CE:** 0 (36 and more), 1 (33–36), 2 (30–33), 3 (27–30), 4 (24–27), 5 (21–24), 6 (15–18), 7 (12–15), 8 (9–12), 9 (6–9), 10 (6 or less)	2.25
Maternal mortality ratio (SDG 3.1.1)	Structural fragility or system failure outcomes	Transformation	Number of maternal deaths per 100,000 live births due to pregnancy-related causes.	Pregnancy risk exposure, quality of obstetric care, access to emergency services, loss of caregivers...	**CE:** 0 (1,000 and more), 1 (900–1,000), 2 (800–900), 3 (700–800), 4 (600–700), 5 (500–600), 6 (400–500), 7 (300–400), 8 (200–300), 9 (70–200), 10 (70 or less)	1.59
<5 Mortality (SDG 3.2.1)	Structural fragility or system failure outcomes	Absorption	Probability of dying before age five per 1,000 live births.	Child survival capacity, effectiveness of primary healthcare, nutrition adequacy, long-term population health	**CE:** 0 (100 and more), 1 (80–100), 2 (70–80), 3 (60–70), 4 (50–60), 5 (40–50), 6 (30–40), 7 (20–30), 8 (10–20), 9 (10 or less), 10 (10 or less)	1.96
Primary Education Completion (Adults)	Capacity and resources	Absorption	Adults (%) aged 25 years and above who completed primary education.	Information access, basic literacy, learning capacity, innovation adoption, livelihood diversification	**CE:** 0(0–41,2%), 1(41,2–60,2%), 2(60,2–74,3%), 3(74,3–85,0%), 4(85,0–88,6%), 5(88,6–92,8%), 6(92,8–96,1%), 7(96,1–98,2%), 8(98,2–99,3%), 9(99,3–99,9%), 10(99,9–100%)	1.57
Food Insecurity Experience (FIES)	Structural fragility or system failure outcomes	Absorption	Individuals (%) experiencing severe food insecurity over the last 12 months, identified by food exhaustion or a full day without eating.	Food access failure, nutritional stress, crisis sensitivity, limited shock absorption, dependence on external support	CE: 0(≥29,2%), 1(20,8–29,2%), 2(13,2–20,8%), 3(8,8–13,2%), 4(6,8–8,8%), 5(5,3–6,8%), 6(3,3–5,3%), 7(2,5–3,3%), 8(1,6–2,5%), 9(1,1–1,6%), 10(0–1,1%)	3.61
Population Without Formal Education	Structural fragility or system failure outcomes	Transformation	Adults (%) reporting no formal education or no completed recognized education level.	Human capital deficit, limited labor opportunities, weak institutional interaction, structural vulnerability	**CE:** 0(≥33,7%), 1(20,1–31,9%), 2(10,5–19,2%), 3(6,9–10,4%), 4(3,6–6,8%), 5(2,7–3,4%), 6(1,9–2,6%), 7(1,1–1,8%), 8(0,7–1,0%), 9(0,3–0,6%), 10(≤0,2%)	2.40

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Suicide mortality rate (SDG 3.4.2)	Structural fragility or system failure outcomes	Absorption	Number of suicide deaths per 100,000 inhabitants recorded over one year.	Psychosocial distress, mental health burden, social isolation, breakdown of support networks, insufficient care access	**CE:** 0(≥16,4), 1(13,8–16,4], 2(10,5–13,8], 3(8,6–10,5], 4(7,7–8,6], 5(6,8–7,7], 6(5,2–6,8], 7(4,5–5,2], 8(3,2–4,5], 9(1,8–3,2], 10(0–1,8]	1.00
Intentional homicide rate (SDG 16.1.1)	Structural fragility or system failure outcomes	Absorption	Number of intentional homicides per 100,000 inhabitants over the last year.	Physical insecurity, violence exposure, erosion of trust, governance weakness, disruption of social cooperation	**CE:** 0(20,5–52,0), 1(11,0–20,5), 2(6,0–11,0), 3(4,0–6,0), 4(3,0–4,0), 5(2,0–3,0), 6(=2,0), 7(1,0–2,0), 8(=1,0), 9(0,0–1,0), 10(=0,0)	2.86
Medical doctors density (SDG 3.c.1)	Capacity and resources	Absorption	Number of medical doctors and specialists per 1,000 inhabitants.	Healthcare availability, epidemic response readiness, avoidable mortality reduction, territorial equity	**CE:** 0(<0,85), 1(0,85–1,7), 2(1,7–2,55), 3(2,55–3,4), 4(3,4–4,25), 5(4,25–5,1), 6(5,1–5,95), 7(5,95–6,8), 8(6,8–7,65), 9(7,65–8,5), 10(>8,5)	2.17
Nurses and Midwives Density (SDG 3.c.1)	Capacity and resources	Absorption	Number of nurses and midwives per 1,000 inhabitants.	Primary care coverage, maternal and child health access, continuity of care, frontline crisis response, protection of vulnerable groups	**CE:** 0(0,0–1,9), 1(1,9–3,8), 2(3,8–5,7), 3(5,7–7,6), 4(7,6–9,5), 5(9,5–11,4), 6(11,4–13,3), 7(13,3–15,2), 8(15,2–17,1), 9(17,1–19,0), 10(19,0–20,9)	2.33
Traditional Health Worker Density	Capacity and resources	Absorption	Number of traditional health workers per 1,000 inhabitants, including traditional midwives and healers.	Local care availability, cultural acceptability, fallback health access, community knowledge systems, care substitution capacity	**CE:** 0(0,0–0,85), 1(0,85–1,7), 2(1,7–2,55), 3(2,55–3,4), 4(3,4–4,25), 5(4,25–5,1), 6(5,1–5,95), 7(5,95–6,8), 8(6,8–7,65), 9(7,65–8,5), 10(8,5–9,4)	1.31
Measles Vaccination Coverage (2nd Dose)	Capacity and resources	Anticipation; Absorption	Children (%) receiving two doses of measles-containing vaccine according to the national schedule.	Epidemic prevention, herd immunity, disease transmission interruption, health cost avoidance, trust in health services	**CE:** 0(≤38,96), 1[38,96–45,07[, 2[45,07–51,17[, 3[51,17–57,27[, 4[57,27–63,38[, 5[63,38–69,48[, 6[69,48–75,59[, 7[75,59–81,69[, 8[81,69–87,79[, 9[87,79–93,90[, 10[93,90–100,00]	2.17
DTP3 Vaccination Coverage	Capacity and resources	Anticipation; Absorption	Children (%) receiving three doses of diphtheria, tetanus and pertussis vaccine.	Protection against vaccine-preventable diseases, child survival, epidemic containment, health system reliability, mortality reduction	**CE :**0(≤67), 1(67–75), 2(75–82), 3(83–87), 4(87–90), 5(90–92), 6(93–94), 7(94–96), 8(96–97), 9(97–98), 10(≥98)	2.17
VII. CULTURAL, IDENTITY-BASED, AND PSYCHOLOGICAL DIMENSION						
Environmental awareness	Adaptive social dynamics	Transformation Anticipation	Likert-scale questionnaire based on concrete proxy questions capturing affective, cognitive, motivational and behavioral dimensions of environmental awareness. Respondents are classified as environmentally aware when their average score reaches at least 4 on a five-point scale. Population-level estimates are derived through weighted aggregation to ensure representativeness across education, residence, age, region and sex.	Pro-environmental behavior, rule compliance, collective resource management	**LN :** 0: 0–5 % | 1: >5–15 % | 2: >15–25 % | 3: >25–35 % | 4: >35–45 % | 5: >45–55 % | 6: >55–65 % | 7: >65–75 % | 8: >75–85 % | 9: >85–95 % | 10: >95–100 %	2.40
Confidence in the future	Adaptive social dynamics	Transformation	Set of binary (0/1) question proxies covering participation in recent elections, engagement in administrative processes, school attendance of children, planting of fruit trees, intention to build or improve housing, participation in collective works of general interest, intention to remain in the community and intention to create or develop a local economic activity.	Long-term planning, proactive behavior, local investment, human capital accumulation, place-based commitment, collective initiative, reduction of short-term coping	**CI:** 0(0) 1(1) 2(2) 3(3) 4(3.5) 5(4) 6(4.5) 7(5) 8(6) 9(7) 10(8)	3.06

Indicator	Concept measured	Core attributes	Metrics (observable variables)	Effect on resilience	Scoring scale (0 worst scenario – 10 best scenario) and class thresholds. Comparative empirical (CE) ; Local normative, linear (LN) ; Composite index (CI)	Weighting
Community belonging and identity	Adaptive social dynamics	AbsorptionTransformation	Set of binary (0/1) question proxies covering the existence of inclusive community spaces, customary rules governing natural resource use, rituals and ceremonies linked to community identity, local rules regulating fishing or land use, codes of conduct governing interaction with nature, transmission of local history and collective memory, shared territorial references and active participation in community activities.	Sense of belonging, social cohesion, trust, compliance with shared rules, intergenerational knowledge transmission, territorial attachment	**CI:** 0(0) 1(1) 2(2) 3(3) 4(3.5) 5(4) 6(4.5) 7(5) 8(6) 9(7) 10(8)	3.42
Traditional knowledge and practices	Adaptive social dynamics	Anticipation Absorption	Set of binary (0/1) question proxies covering knowledge of medicinal plants, food conservation techniques, traditional agricultural and fishing practices, anticipation of climatic and environmental hazards, navigation and environmental observation skills, construction techniques using local materials and the existence and transmission of artisanal know-how.	Local ecological knowledge, sustainable resource use, environmental adaptation, food security practices, health autonomy, intergenerational learning	**CI:** 0(0) 1(1) 2(2) 3(3) 4(3.5) 5(4) 6(4.5) 7(5) 8(5.5) 9(6) 10(7)	2.56
Adaptation versus resignation	Adaptive social dynamics	Transformation	Set of binary (0/1) question proxies covering perceived ability to act following financial losses, crop diseases, climate hazards, theft or poor harvests, as well as willingness to adapt practices or relocate to improve living conditions. Resignation-oriented responses are inversely coded.	Agency perception, problem-solving behavior, anticipation of risks, learning capacity, psychological flexibility, willingness to change strategies	**CI:** 0(0) 1(1) 2(2) 3(2.5) 4(3) 5(3.5) 6(4) 7(4.5) 8(5) 9(5.5) 10(6)	2.10
Cohesion, mutual aid and collaboration	Adaptive social dynamics	Absorption ; Transformation	Set of binary (0/1) question proxies covering receipt and provision of labor, food or assistance within the community, reliance on support outside the household, participation in mutual aid or collective labor groups, involvement in collective arrangements for schooling or care, contribution to community activities and collective problem-solving.	Mutual support, resource sharing, collective problem-solving, trust networks, coordinated crisis response, community solidarity	**CI:** 0(0) 1(1) 2(2) 3(3) 4(3.5) 5(4) 6(4.5) 7(5) 8(6) 9(7) 10(8)	2.72
Awareness of rights and civic participation	Adaptive social dynamics	Transformation	Set of binary (0/1) question proxies covering participation in civic actions, knowledge of fundamental rights, awareness of legal mechanisms, knowledge of institutions providing legal support, experience with filing complaints or claims and interaction with rights-based organizations.	Empowerment, access to justice, accountability mechanisms, participation in decision-making, institutional engagement, protection against abuse	**CI:** 0(0) 1(1) 2(2) 3(2.5) 4(3) 5(3.5) 6(4) 7(4.5) 8(5) 9(5.5) 10(6)	1.82
Displacement Reception and Integration	Adaptive social dynamics	Absorption	Set of binary (0/1) question proxies covering perceived welcome of newly arrived people, access to social networks, absence of discrimination (inversely coded), fair treatment by institutions, hosting of displaced persons, access to economic support and access to basic and education services.	Social inclusion, tension reduction, economic integration, equitable access to services, conflict prevention, community stability	**CI:** 0(0) 1(1) 2(2) 3(3) 4(3.5) 5(4) 6(4.5) 7(5) 8(6) 9(7) 10(8)	2.10
Spiritual grounding and beliefs	Adaptive social dynamics	Absorption ; Transformation	Set of binary (0/1) question proxies covering the role of spiritual and religious practices in psychological coping, solidarity, peaceful coexistence between beliefs, respect for nature, coping with adversity and rejection of fatalism or passivity (inversely coded).	Meaning-making, emotional regulation, hope and perseverance, moral norms, solidarity values, coping with loss and uncertainty, respect for nature	**CI:** 0(0) 1(1) 2(2) 3(2.5) 4(3) 5(3.5) 6(4) 7(4.5) 8(5) 9(5.5) 10(6)	1.82

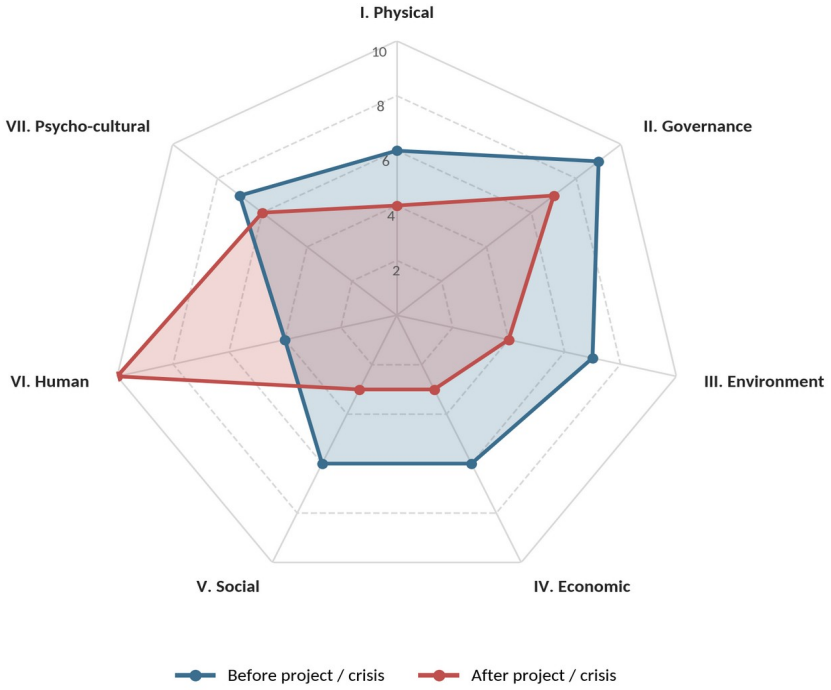
Annex 3. Illustrative example of quantile-based discretization: safely managed sanitation services



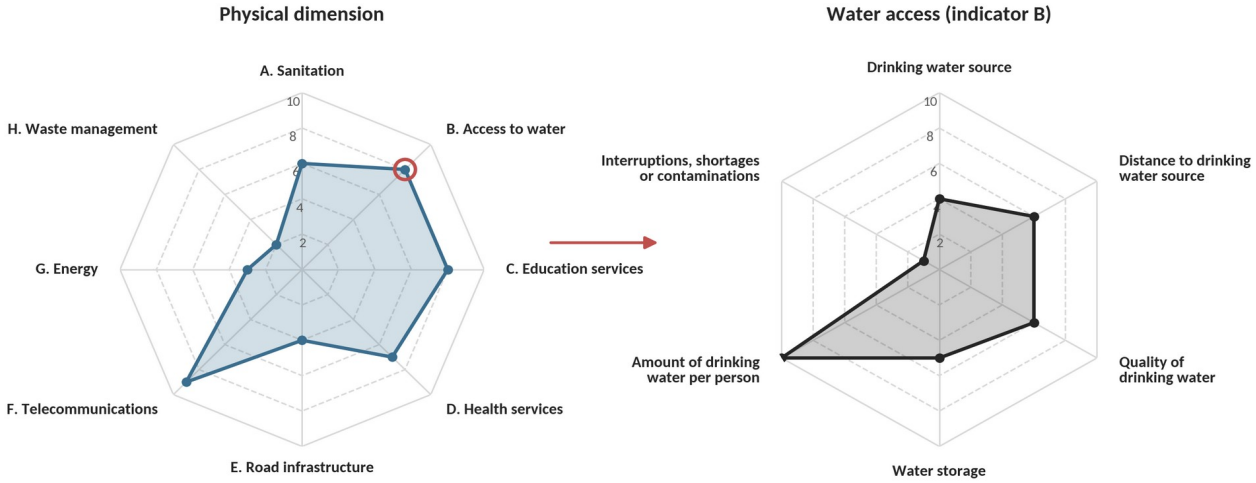
Source: Author.

Annex 4. Radar-based visualization of resilience scores

a) Multidimensional resilience profile (seven dimensions)

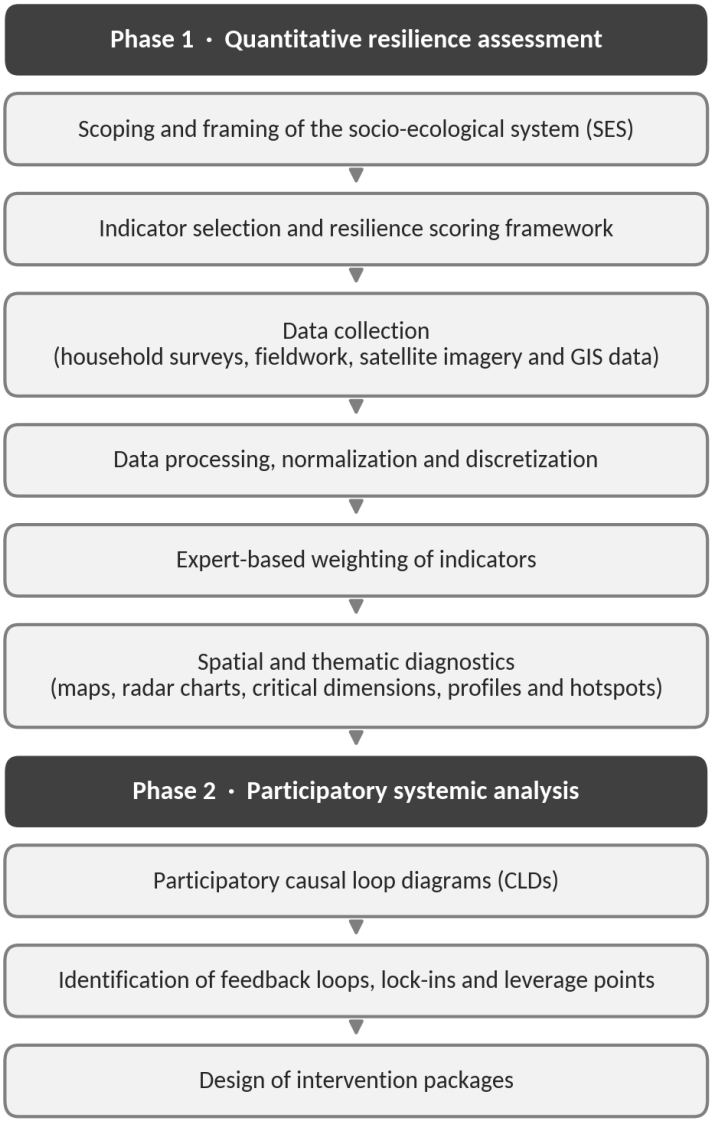


b) Disaggregated dimensional scores (physical dimension and water access)



Source: Author, based on Sayer et al. (2017) and FAO (2019). Illustrative profiles; scores on the 0–10 scale.

Annex 5. Sequential workflow of the IRLA approach



Source: Author.